Supplementary Information for:

Pathway Selection between Click and Acyl Transfer Reactions Driven by Aminoacyl Phosphates

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1. Methods

UPLC methods.

Method A

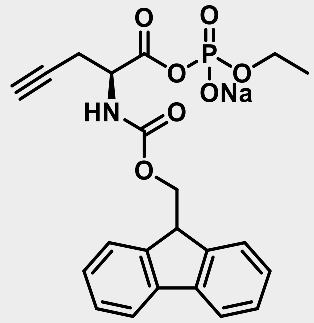
Time (min)	Water (%)	Acetonitrile (%)
0	95	5
3	50	50
15	10	90
16	10	90
17	10	90
17.5	95	5
20	95	5

Method B

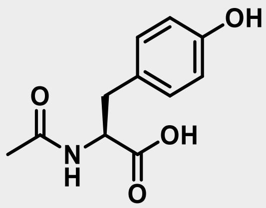
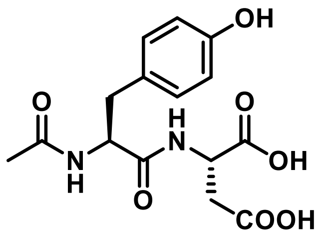
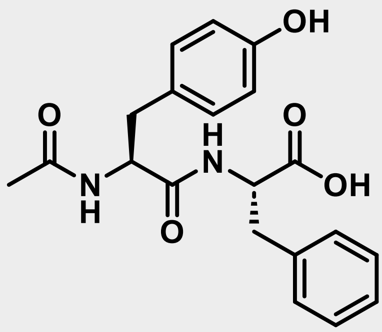
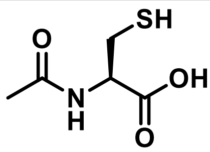
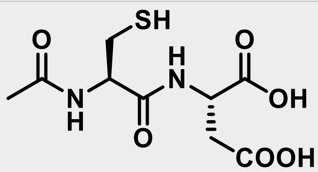
Time (min)	Water (%)	Acetonitrile (%)
0	95	5
15	10	90
16	10	90
17	10	90
17.5	95	5
20	95	5

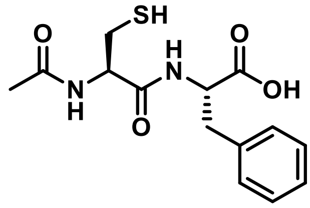
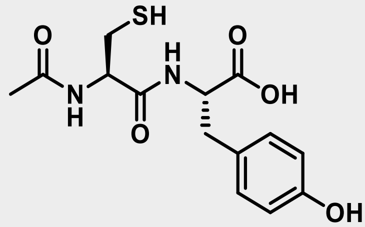
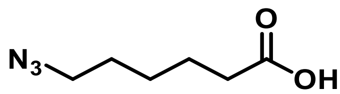
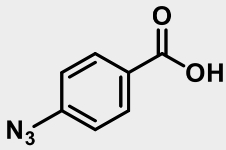
Method B was used for libraries containing the dipeptides **Ac-CF** and **Ac-CD**. For the rest of the libraries, Method A was used.

Supplementary Table 1: Name, chemical structure of aminoacyl phosphate ester and its retention time determined from UPLC measurements.

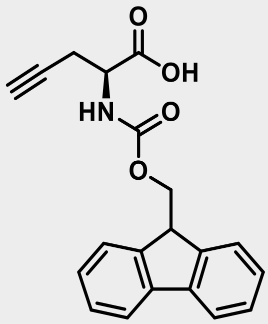
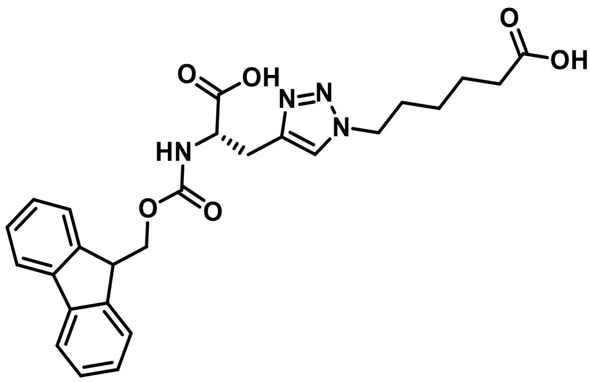
Name	Structure
Fmoc-Alkyne-EP (1-EP) (Retention time = 10.8 min)	

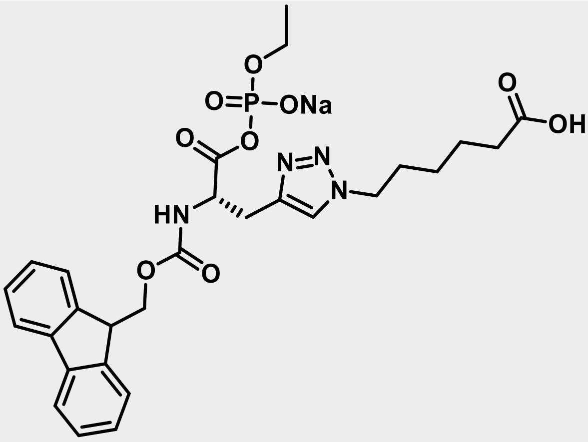
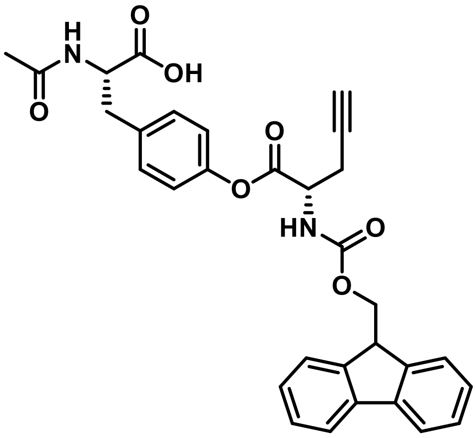
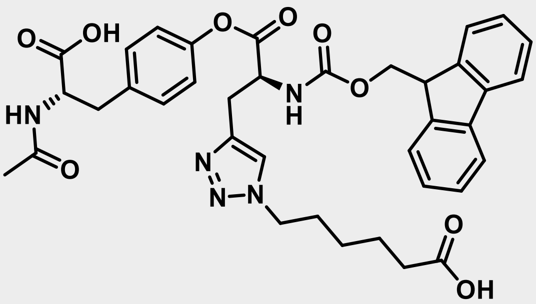
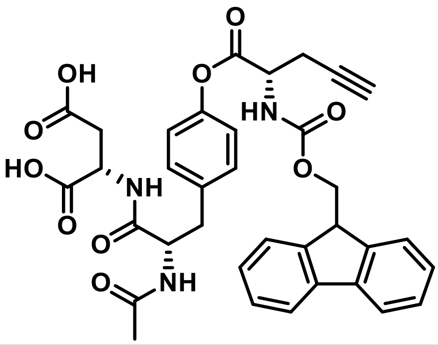
Supplementary Table 2: Abbreviations and chemical structures of peptide sequences and azides used in this work.

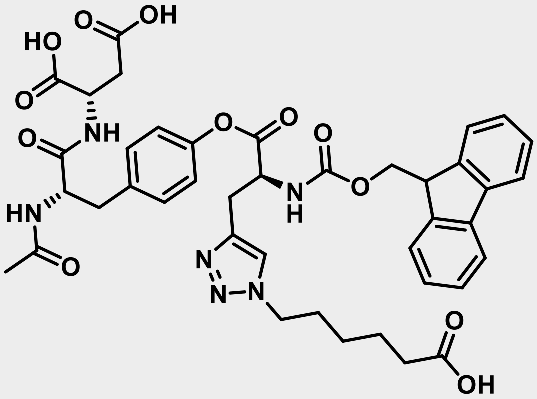
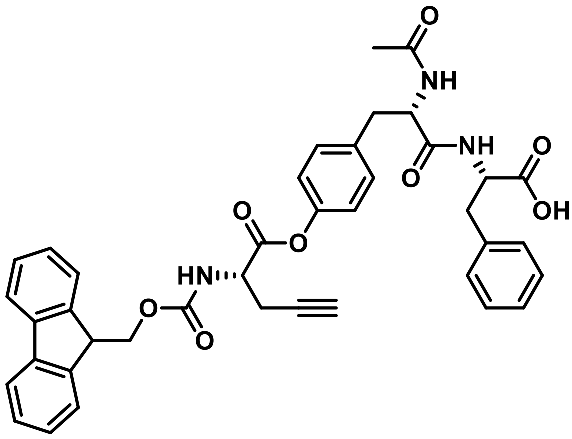
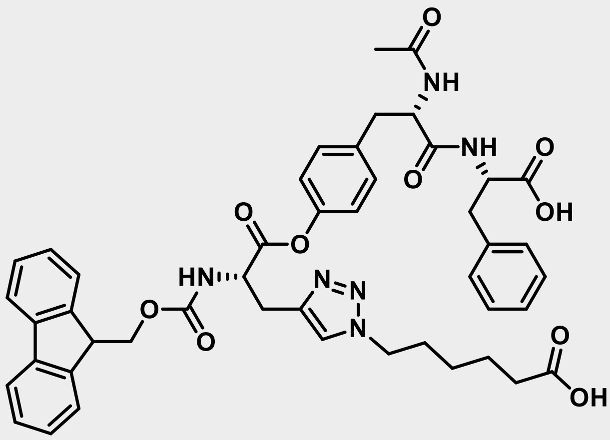
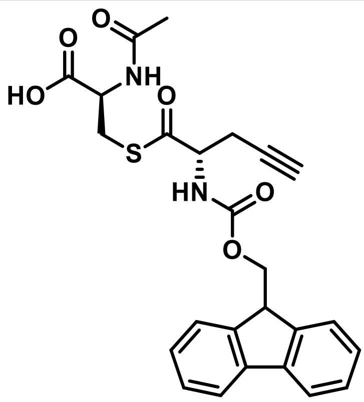
Name	Structure
Ac-Y(a)	
Ac-YD(b)	
Ac-YF(c)	
Ac-C(a′)	
Ac-CD(b′)	

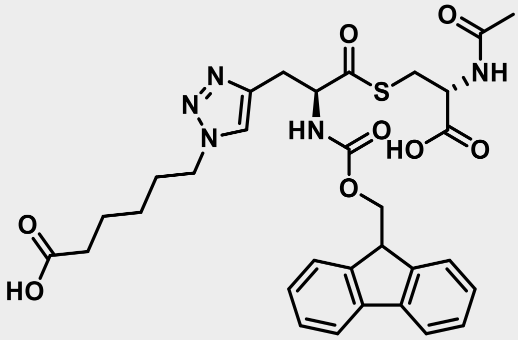
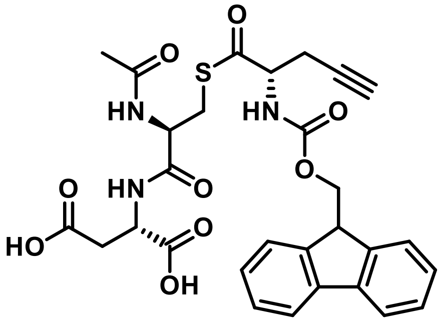
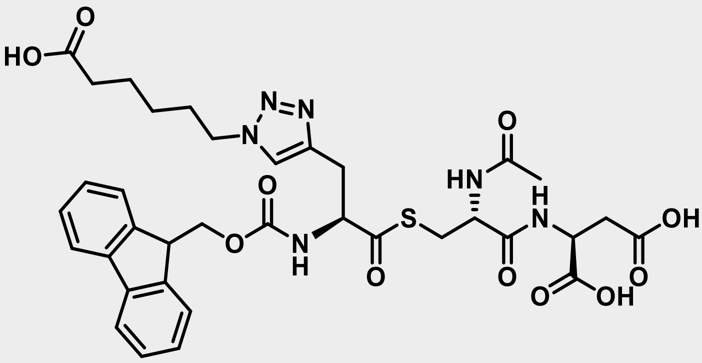
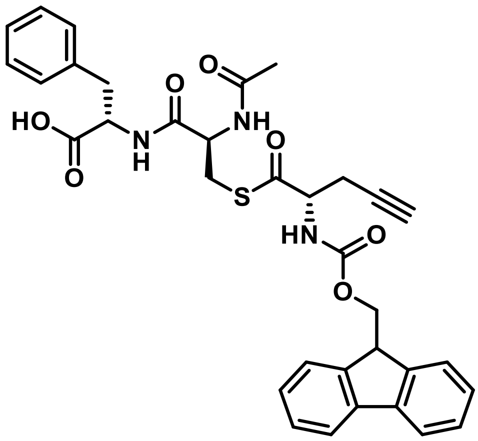
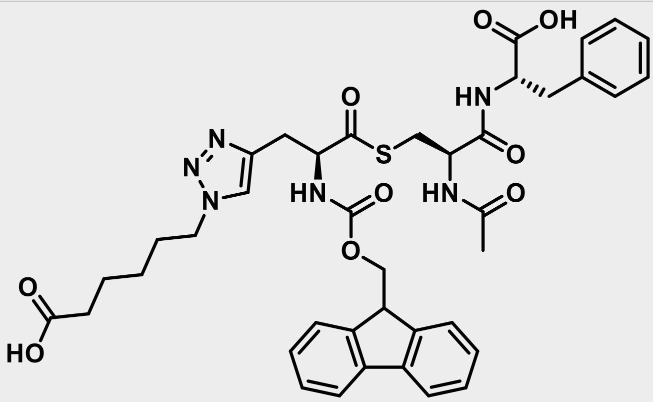
Ac-CF(c′)	
Ac-CY	
azide (6-Azidohexanoic acid)	
aromatic azide (4-azidobenzoic acid)	

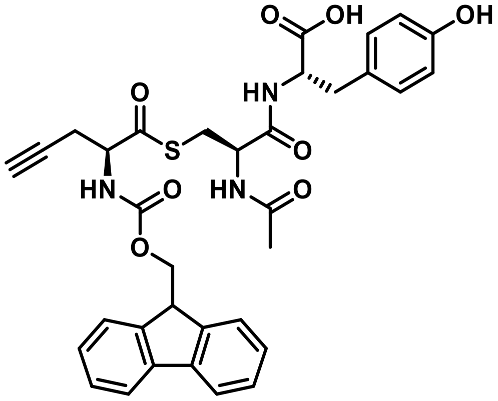
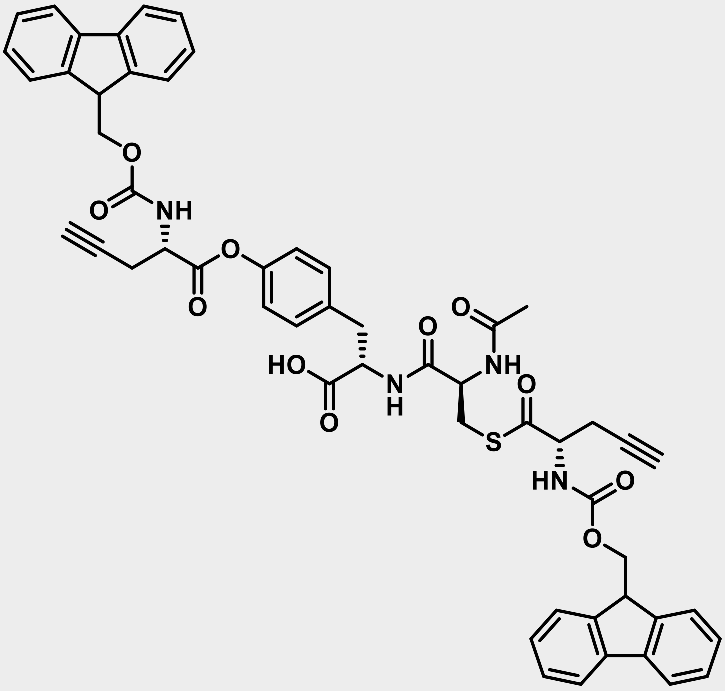
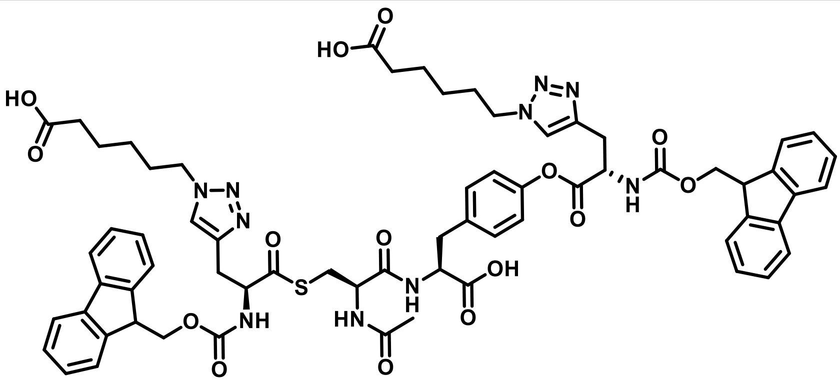
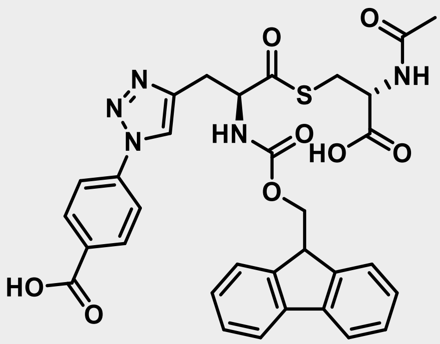
Supplementary Table 3: Abbreviations used and retention times determined by UPLC for different molecules formed in the libraries.

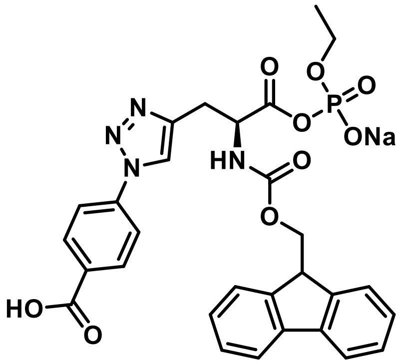
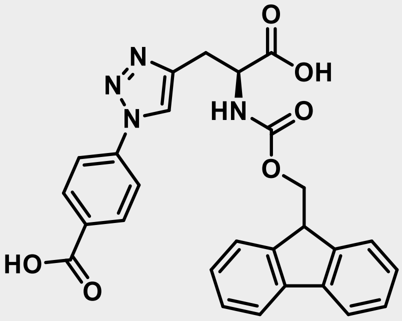
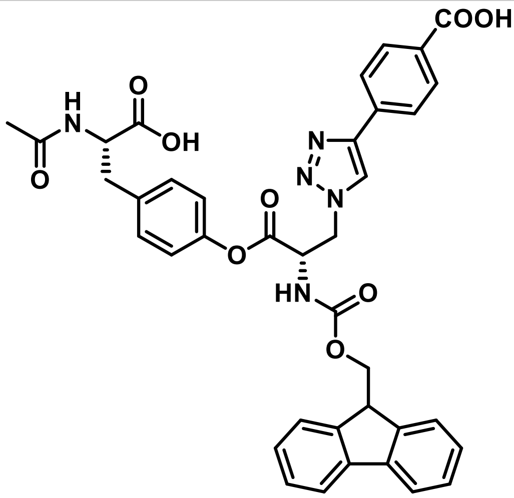
Name	Structure
Fmoc-Alkyne (1) (Retention time = 7.4 min)	
Click (2) (Retention time = 6.0 min)	

Click-EP (2-EP) (Retention time = 7.6 min)	
Ac-Y-Ester (3a)	
Ac-Y-Ester-click (4a) (Retention time = 6.4 min)	
Ac-YD-Ester (3b) (Retention time = 6.8 min)	

Ac-YD-Ester-click (4b) (Retention time = 5.9 min)	
Ac-YF-Ester (3c) (Retention time = 8.7 min)	
Ac-YF-Ester-click (4c) (Retention time = 7.3 min)	
Ac-C-Thioester (5a´) (Retention time = 6.8 min)	

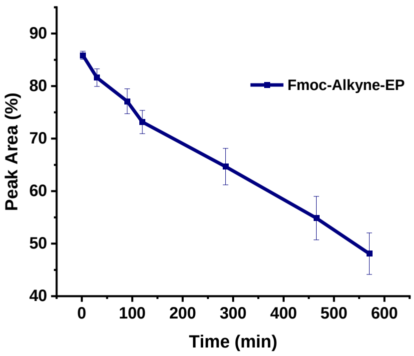
Ac-C-Thio-click (6a´) (Retention time = 5.8 min)	
Ac-CD-Thioester (5b´) (Retention time = 10.3 min)	
Ac-CD-Thio-click (6b´) (Retention time = 9.5 min)	
Ac-CF-Thioester (5c´) (Retention time = 12.1 min)	
Ac-CF-Thio-click (6c´) (Retention time = 11.1 min)	

Ac-CY-Thioester (7) (Retention time = 6.8 min)	
Ac-CY-Diester (8) (Retention time = 11.6 min)	
Ac-CY-Diester-click (9) (Retention time = 5.9 min)	
Ac-C-Thio-click-aromatic (10a´) (Retention time = 6.3 min)	

Click-aromatic-EP (11-EP) (Retention time = 9.5 min)	
Click-aromatic (11) (Retention time = 6.6 min)	
Ac-Y-Ester-click-aromatic (12a) (Retention time = 7.1 min)	

Half-life of Fmoc-aminoacyl phosphate ester calculated by using equation: $t_{1/2} = 0.693/k$, assuming first-order kinetics.

Molecule	Half-life
Fmoc-Alkyne-EP (1-EP)	11.55 h



2. Synthesis and characterization of compounds

2.1 General Synthesis

1-EP was synthesized by following the procedure given below[1] :

i) **General procedure for the synthesis of Bis (tetraethylammonium salt) ethyl phosphate:**

Ethyl phosphorodichloridate was added dropwise to a ten-fold excess of distilled water in an ice-cooled round-bottom flask. After stirring the reaction mixture for one-hour, hydrochloric acid was generated as a by-product and was removed by rotary evaporation. This yielded ethyl phosphoric acid as a clear oil. The resulting ethyl phosphoric acid was neutralized with tetraethylammonium hydroxide (35% w/w solution, 2.0 eq.), resulting in a white paste. The paste was obtained by freeze-drying and used for coupling reactions without further purification.

ii) **General procedure for the synthesis of Fmoc-aminoacyl ethyl phosphate:**

Fmoc-L-amino acid (3.75 mmol, 1.0 eq) was dissolved in dry DCM (30 mL). DCC (3.75 mmol, 1.0 eq) was added to this solution and then stirred for 3-5 minutes which resulted in a white precipitate. Bis(tetraethylammonium salt) ethyl phosphate (3.75 mmol, 1.2 eq), pre-dissolved in dry DCM (10 mL), was added, and stirred for 1 hour at room temperature. The progress of the reaction was monitored using TLC analysis [MeOH: DCM (10:90)]. The reaction mixture was then filtered to remove the white DCU precipitate. The filtrate was evaporated under reduced pressure and the residue was purified by automated RP flash column chromatography (C18-AQ, H_2O/Acetonitrile). Fractions containing the product were collected and lyophilized to obtain the aminoacyl ethyl phosphate as TEA-salt.

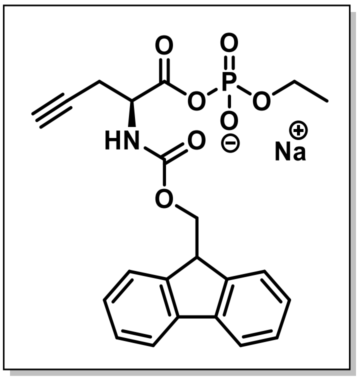
iii) **General procedure for ion exchange:**

The TEA salt of Fmoc-aminoacyl ethyl phosphate was dissolved in a minimum amount of water and then passed through a column which contained Dowex® 50WX8 ion exchange resin loaded with Na^+ as a counter ion. The collected water solution was then frozen in liquid nitro gen and lyophilized to dryness. This procedure yields the sodium salt of Fmoc-aminoacyl ethyl phosphate as a white hygroscopic solid. The final compound (1-EP) was isolated with 96% purity. Due to their hygroscopic nature, we observed that the stability of ethyl phosphate esters is greater at -80 °C than at -20 °C.

2.2 Characterization of compounds

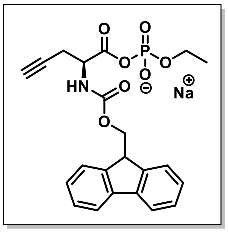
2.2.1 Characterization of 1-EP by NMR

i) (*S*)-2-((((9H-fluoren-9-yl)methoxy)carbonyl)amino)-pent-4-ynoic(ethyl phosphoric anhydride) (Sodium Salt):

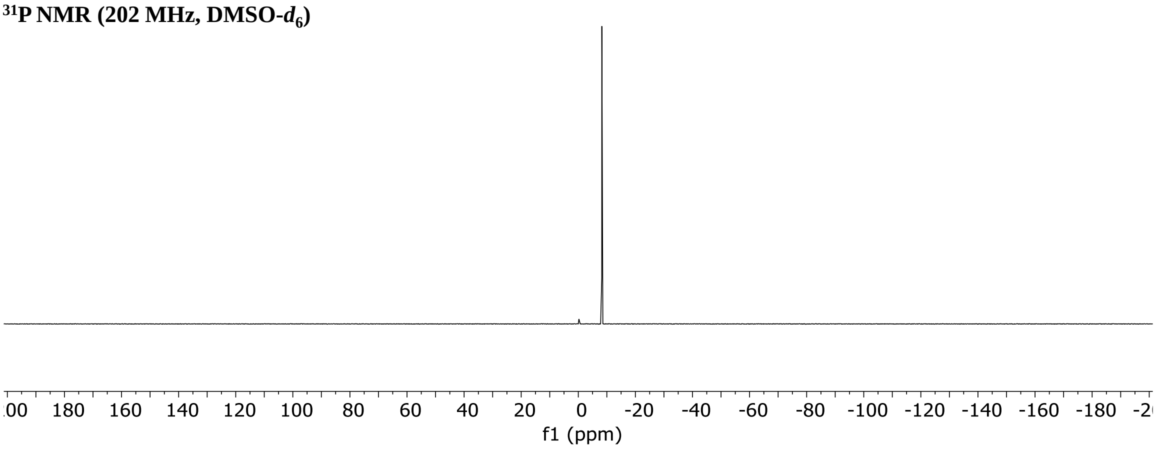


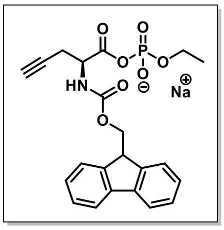
The title compound was synthesized as white fluffy solid in 46% yield.
^{31}P NMR (202 MHz, DMSO-*d*$_6$) δ -8.24.
^{1}H NMR (500 MHz, DMSO-*d*$_6$) δ 7.89 (d, *J* = 7.5 Hz, 2H), 7.81 (d, *J* = 8.7 Hz, 1H), 7.74 (d, *J* = 7.5 Hz, 2H), 7.42 (td, *J* = 7.5, 1.2 Hz, 2H), 7.34 (ddd, *J* = 7.4, 6.7, 1.3 Hz, 2H), 4.33 – 4.22 (m, 3H), 4.20 –

4.15 (m, 1H), 3.76 (p, *J* = 7.2 Hz, 2H), 2.88 (t, *J* = 2.6 Hz, 1H), 2.68 (ddd, *J* = 16.8, 4.5, 2.7 Hz, 1H), 2.55 (ddd, *J* = 16.8, 9.7, 2.6 Hz, 1H), 1.07 (t, *J* = 7.1 Hz, 3H).

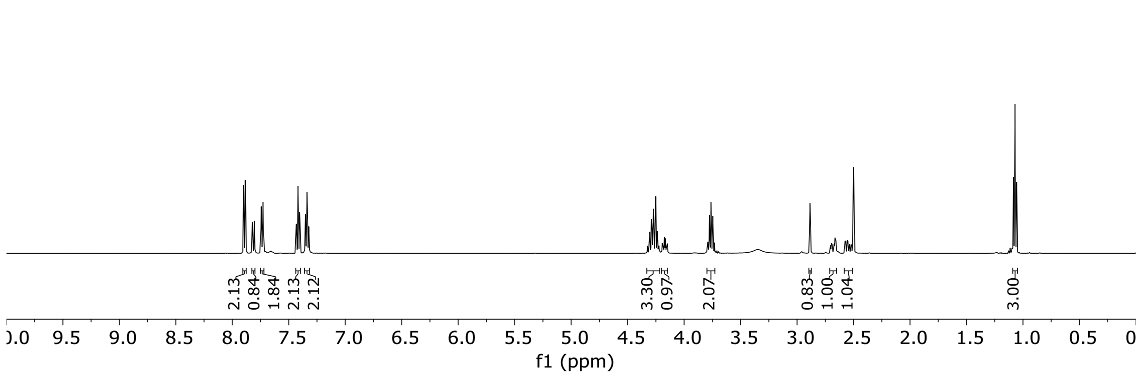


^{31}P NMR (202 MHz, DMSO-d_6)

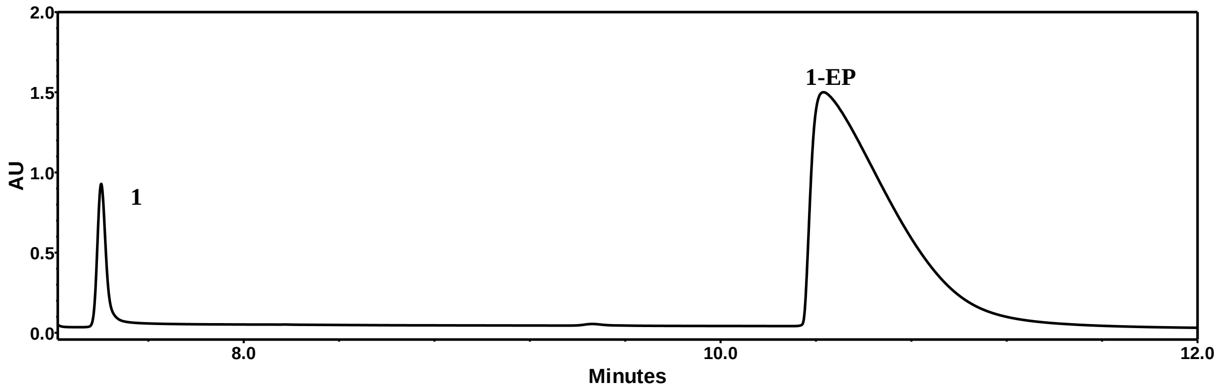




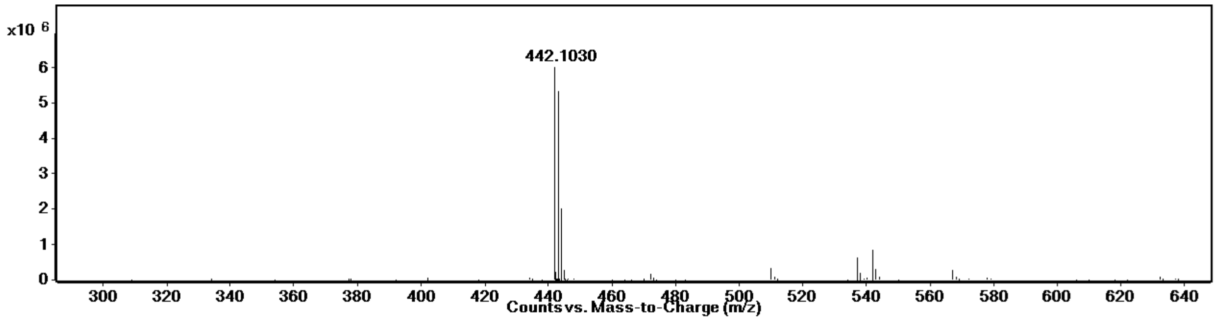
^{1}H NMR (500 MHz, DMSO-d_6)



2.2.2 Characterization of 1-EP by UPLC and UPLC-MS

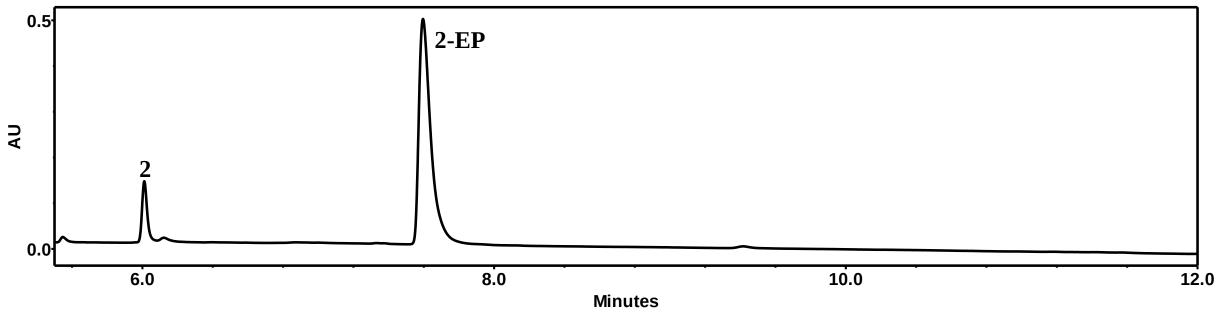


UPLC chromatogram of **1-EP. [*(S)*-2-((((9H-fluoren-9-yl) methoxy) carbonyl) amino)-3-azidopropanoic (ethyl phosphoric anhydride)]**

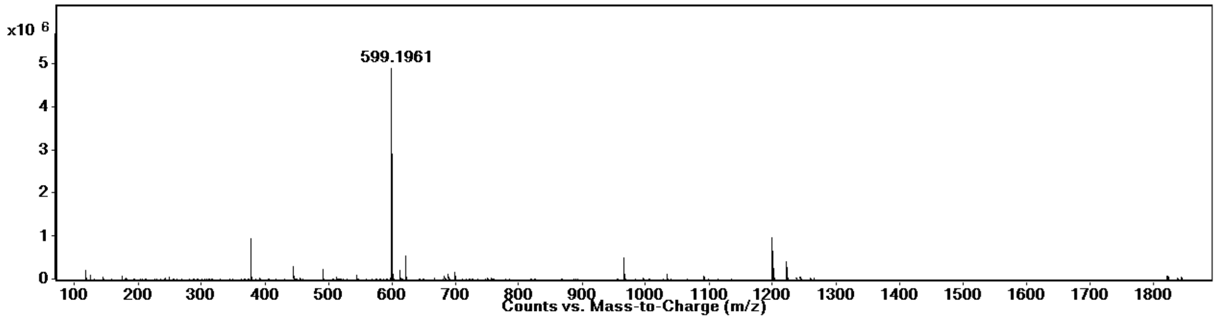


Mass spectra of **1-EP [*(S)*-2-((((9H-fluoren-9-yl) methoxy) carbonyl) amino)-3-azidopropanoic (ethyl phosphoric anhydride)]:** Calculated m/z [M-H]$^-$: 442.1061, Observed m/z [M-H]$^-$: 442.1030

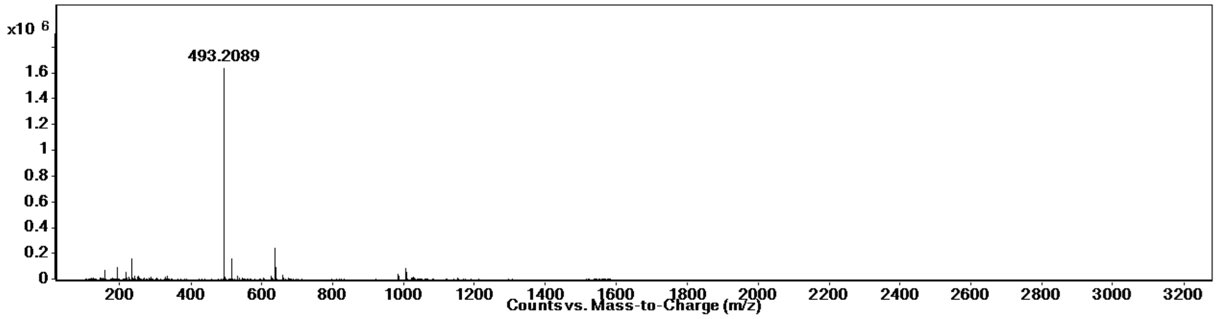
2.2.3 Characterization of Esters, Thioesters and their corresponding click products by UPLC and UPLC-MS



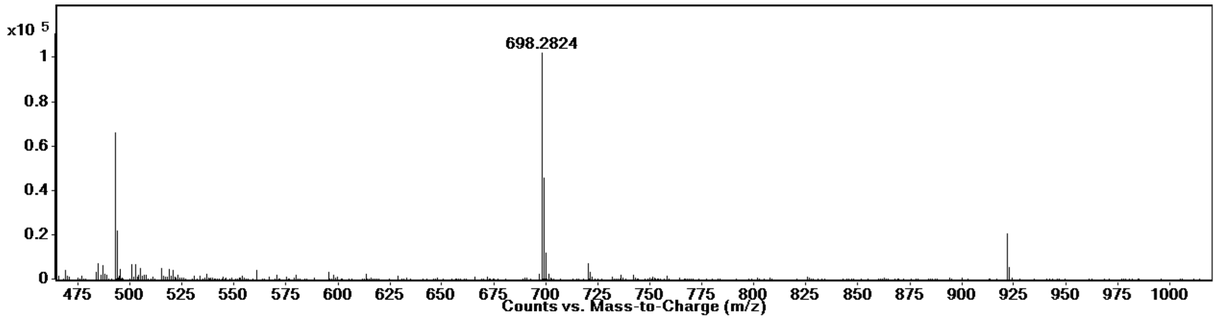
UPLC chromatogram of **2-EP**



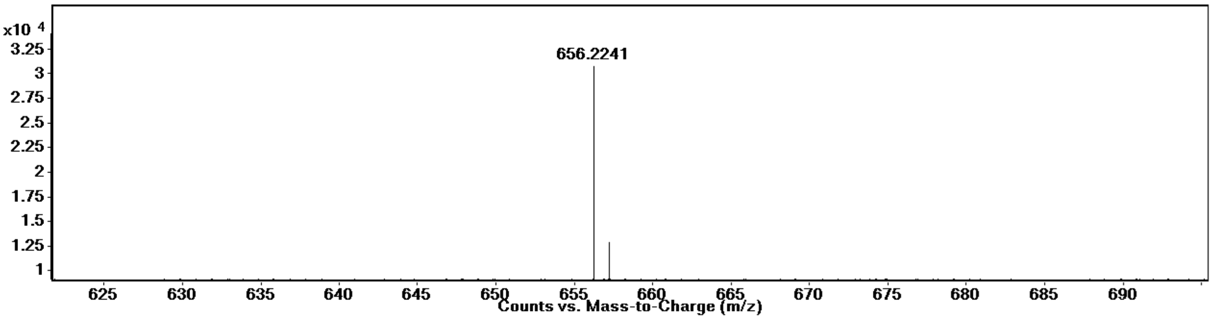
Mass spectra of **2-EP**: Calculated m/z [M-H]$^-$: 599.1912, Observed m/z [M-H]$^-$: 599.1961



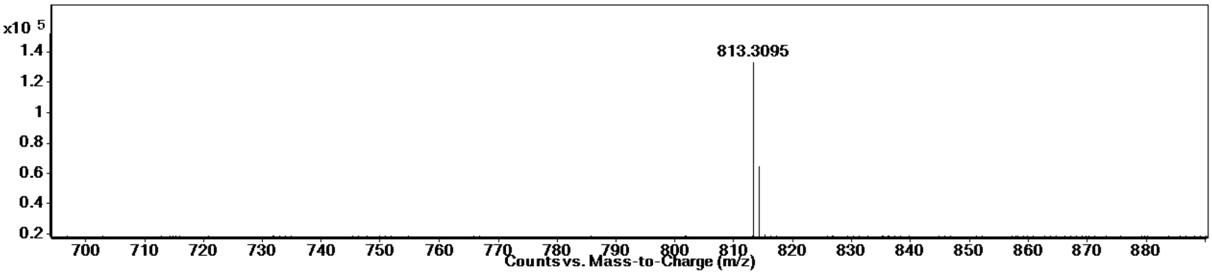
Mass spectra of **2** : Calculated m/z [M+H]$^+$: 493.2082, Observed m/z [M+H]$^+$: 493.2089



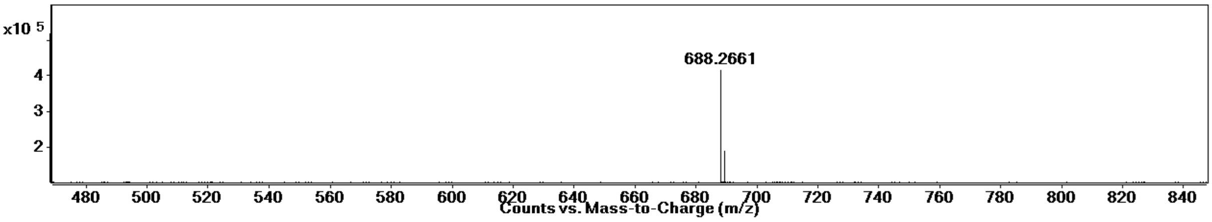
Mass spectra of **4a**: Calculated m/z [M+H]$^+$: 698.2821, Observed m/z [M+H]$^+$: 698.2824



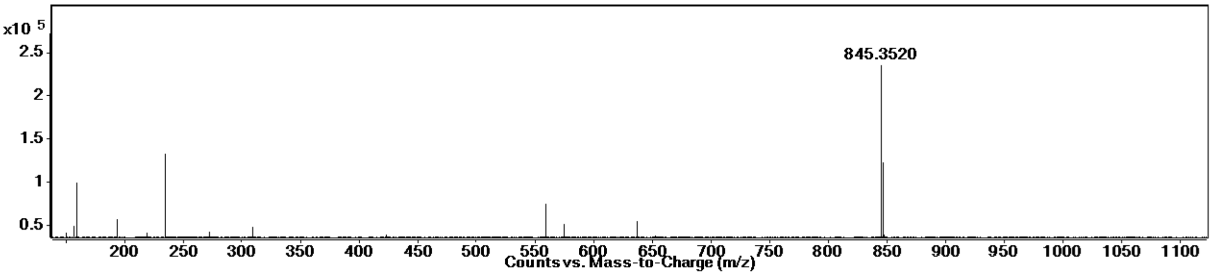
Mass spectra of **3b**: Calculated m/z [M+H]$^+$: 656.2239, Observed m/z [M+H]$^+$: 656.2241



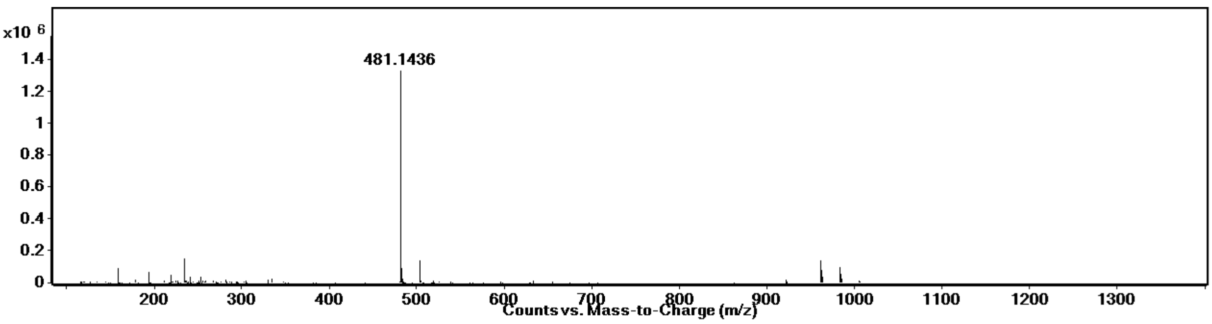
Mass spectra of **4b**: Calculated m/z [M+H]$^+$: 813.3090, Observed m/z [M+H]$^+$: 813.3095



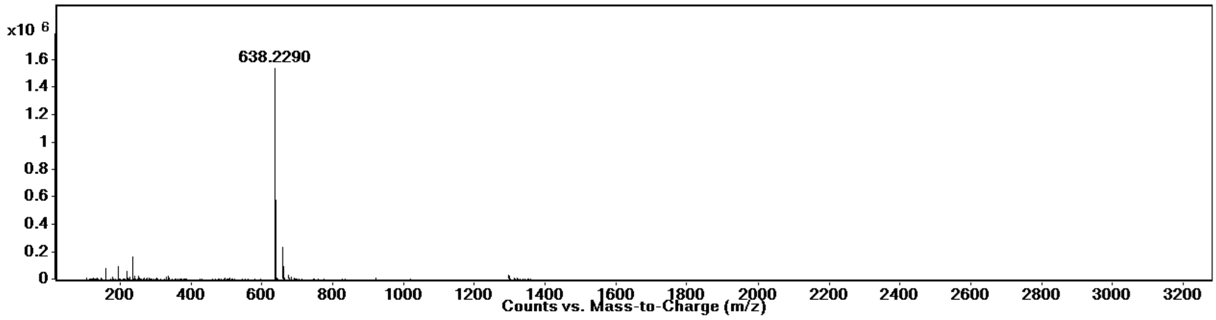
Mass spectra of **3c**: Calculated m/z [M+H]$^+$: 688.2653, Observed m/z [M+H]$^+$: 688.2661



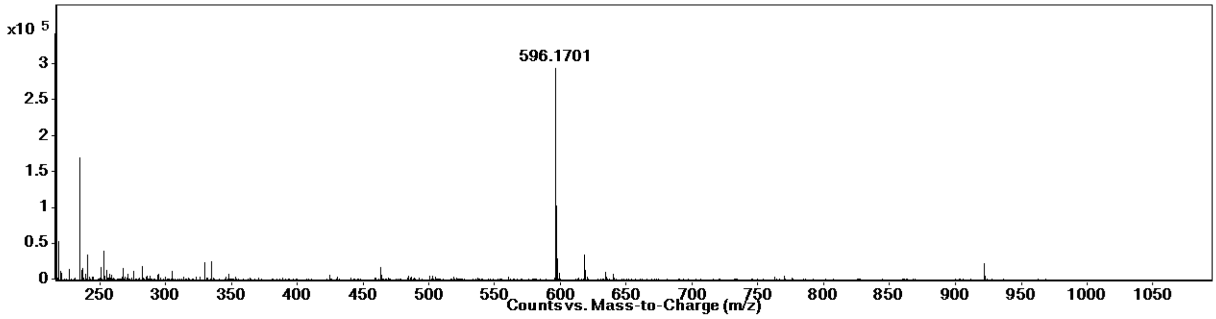
Mass spectra of **4c**: Calculated m/z [M+H]$^+$: 845.3505, Observed m/z [M+H]$^+$: 845.3520



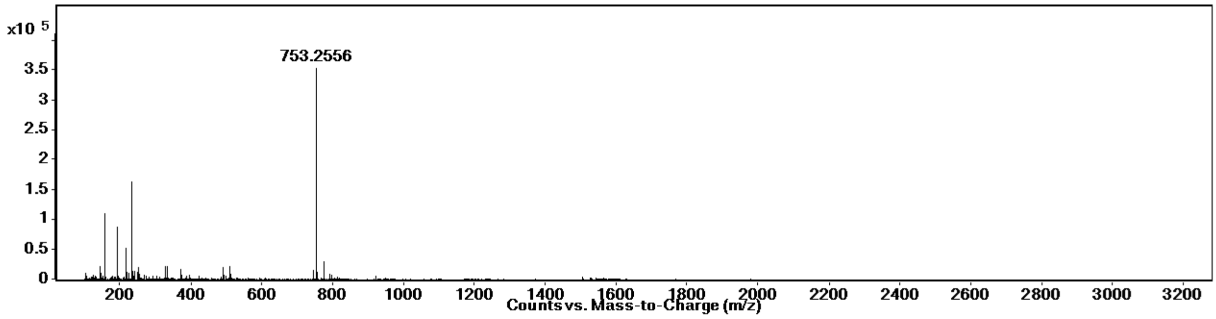
Mass spectra of **5a´**: Calculated m/z [M+H]$^+$: 481.1428, Observed m/z [M+H]$^+$: 481.1436



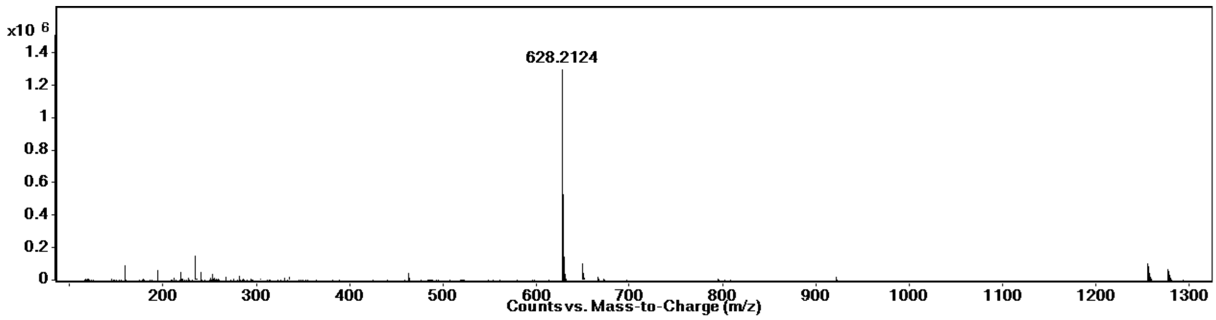
Mass spectra of **6a**´: Calculated m/z [M+H]+: 638.2279, Observed m/z [M+H]+: 638.2290



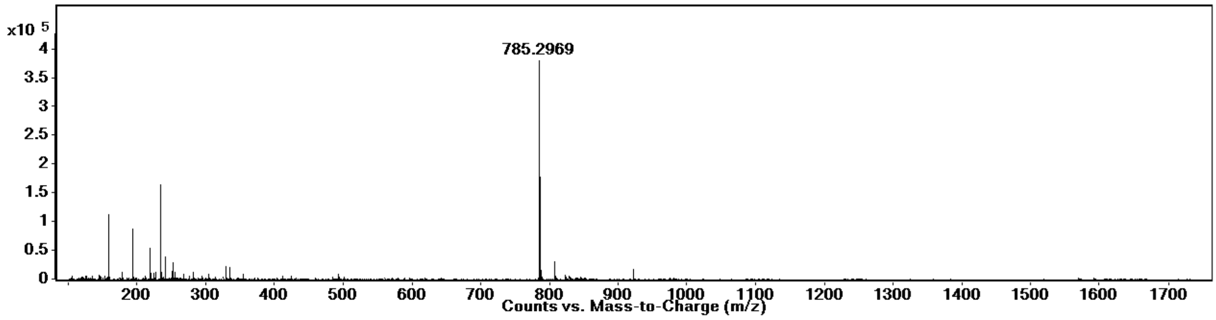
Mass spectra of **5b**´: Calculated m/z [M+H]+: 596.1697, Observed m/z [M+H]+: 596.1701



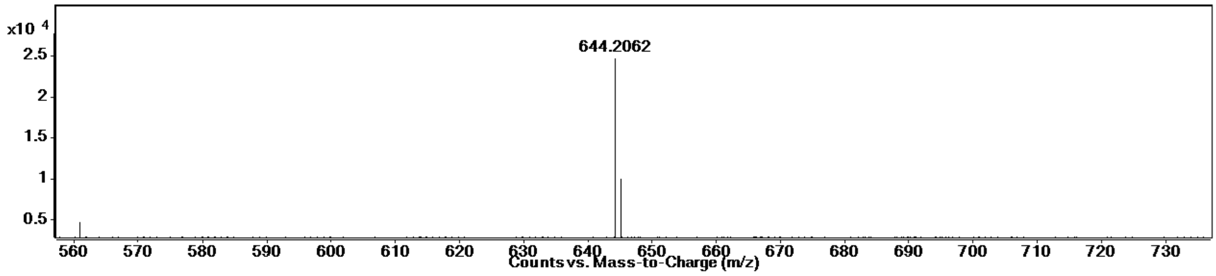
Mass spectra of **6b**´: Calculated m/z [M+H]+: 753.2549, Observed m/z [M+H]+: 753.2556



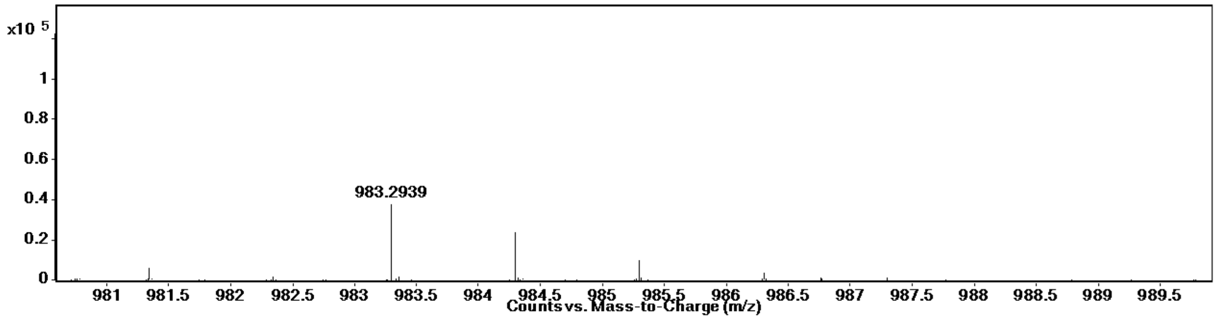
Mass spectra of **5c**´: Calculated m/z [M+H]+: 628.2112, Observed m/z [M+H]+: 628.2124



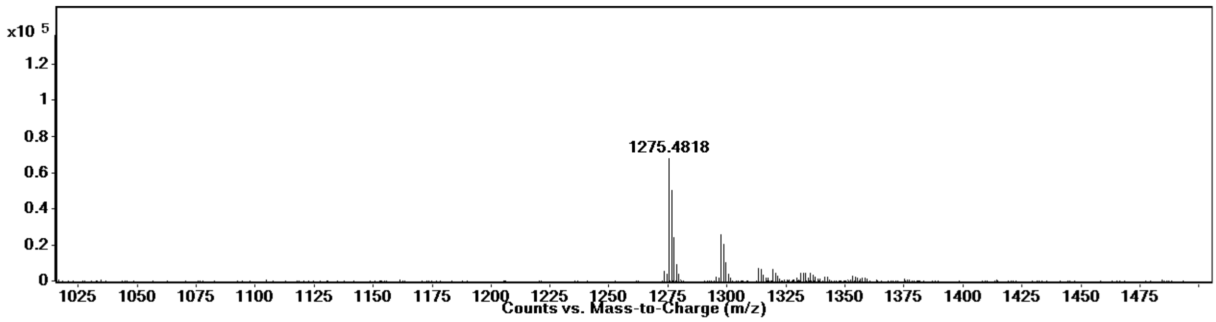
Mass spectra of **6c´**: Calculated m/z [M+H]$^+$: 785.2963, Observed m/z [M+H]$^+$: 785.2969



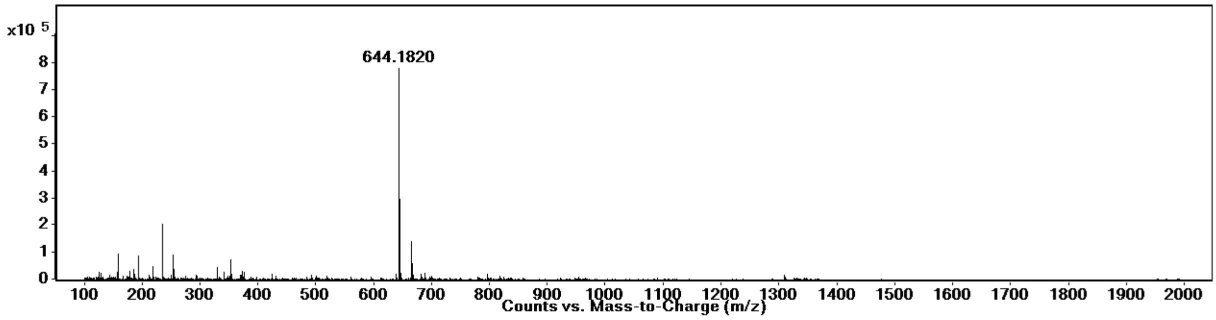
Mass spectra of **7**: Calculated m/z [M+H]$^+$: 644.2061, Observed m/z [M+H]$^+$: 644.2062



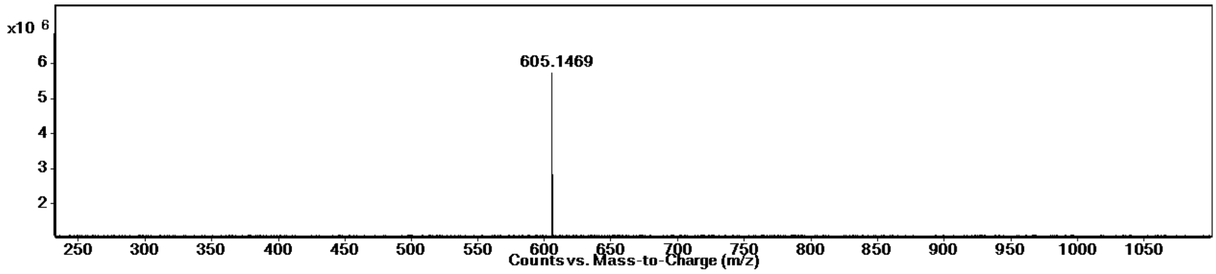
Mass spectra of **8**: Calculated m/z [M+Na]$^+$: 983.2933, Observed m/z [M+Na]$^+$: 983.2939



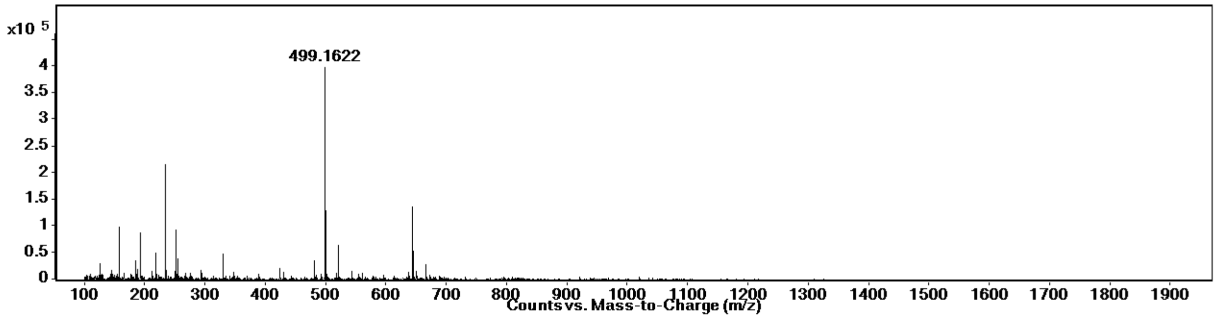
Mass spectra of **9**: Calculated m/z [M+H]$^+$: 1275.4816, Observed m/z [M+H]$^+$: 1275.4818



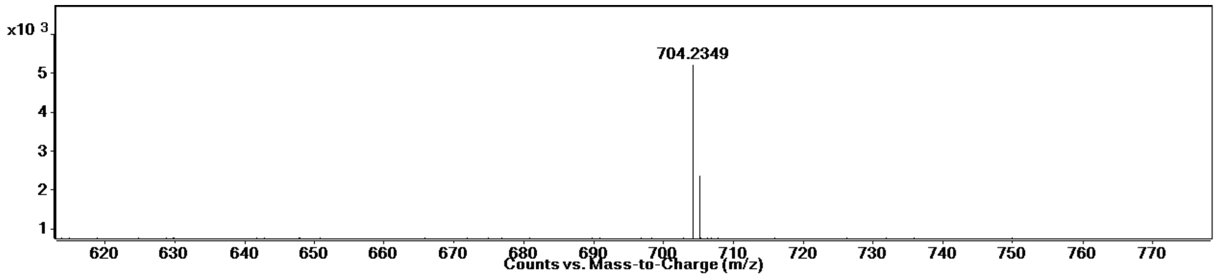
Mass spectra of **10a´:** Calculated m/z [M+H]⁺: 644.1810, Observed m/z [M+H]⁺: 644.1820



Mass spectra of **11-EP**: Calculated m/z [M+H]⁺: 605.1443, Observed m/z [M+H]⁺: 605.1469



Mass spectra of **11**: Calculated m/z [M+H]⁺: 499.1612, Observed m/z [M+H]⁺: 499.1622

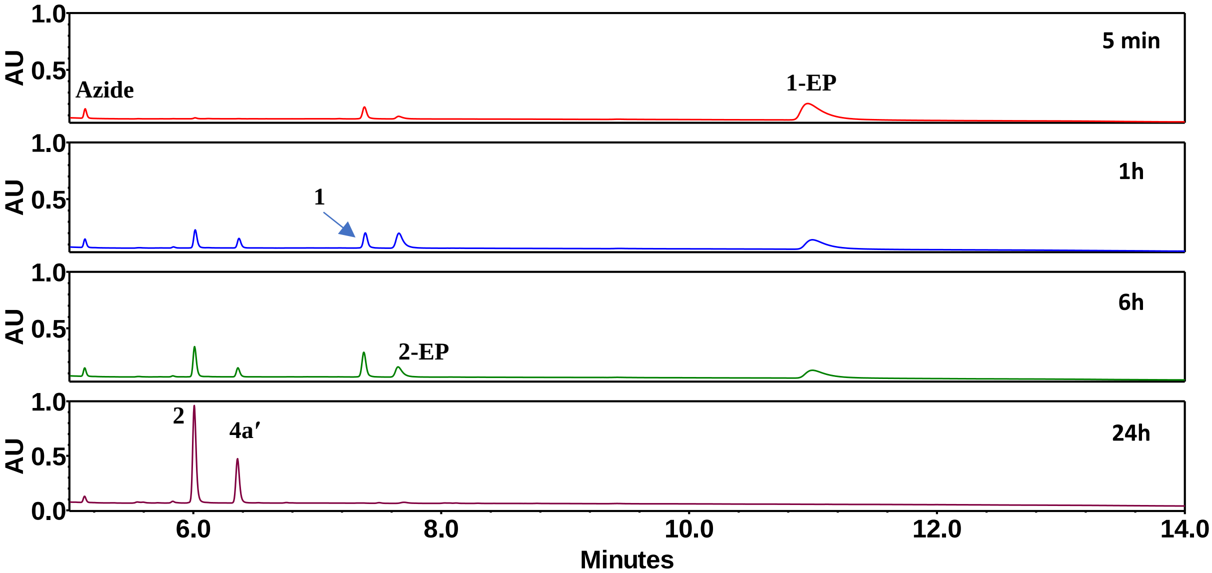


Mass spectra of **12a´**: Calculated m/z [M+H]⁺: 704.2351, Observed m/z [M+H]⁺: 704.2349

3. Supplementary figures

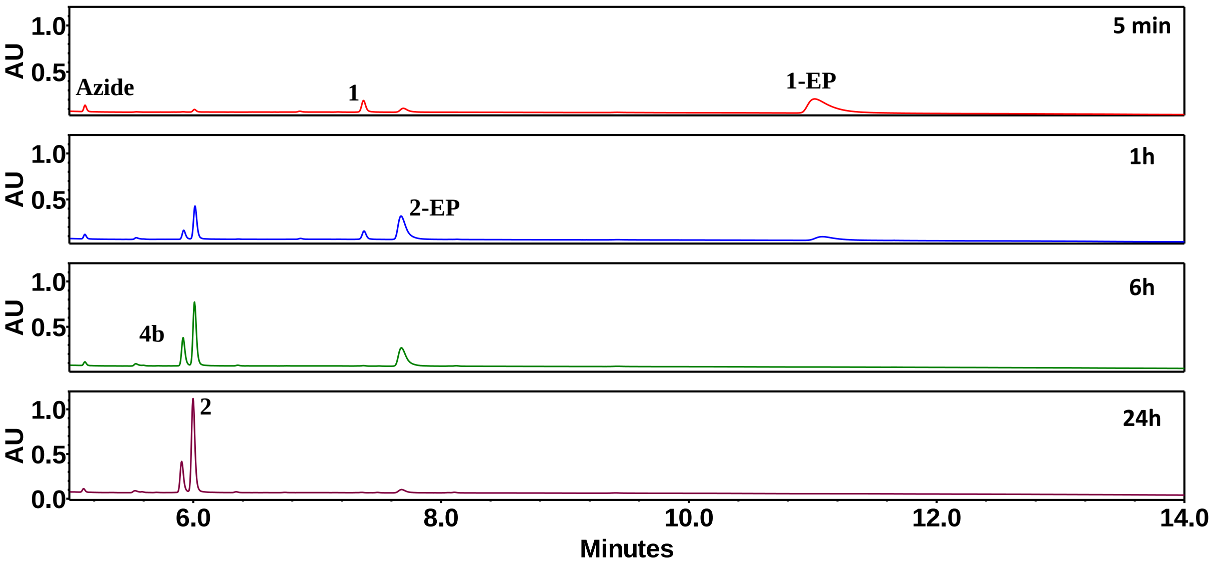
3.1 Libraries from tyrosine derivatives

3.1.1 Ac-Y



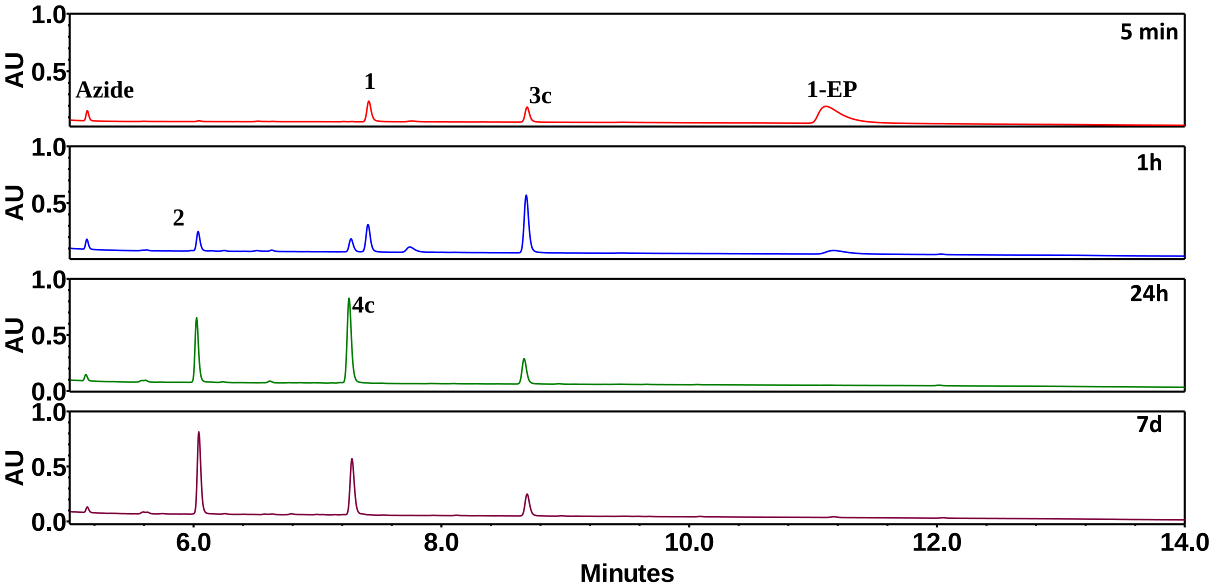
Supplementary Fig. 1. Time-dependent UPLC chromatograms of reaction between 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-Y** in 0.2 M bicarbonate buffer (pH 10.0).

3.1.2 Ac-YD

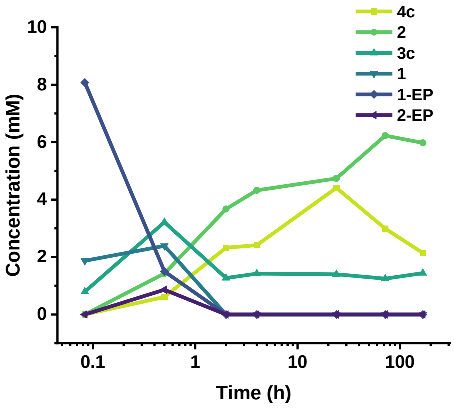


Supplementary Fig. 2. Time-dependent UPLC chromatograms of reaction between 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-YD** in 0.2 M bicarbonate buffer (pH 10.0).

3.1.3 Ac-YF



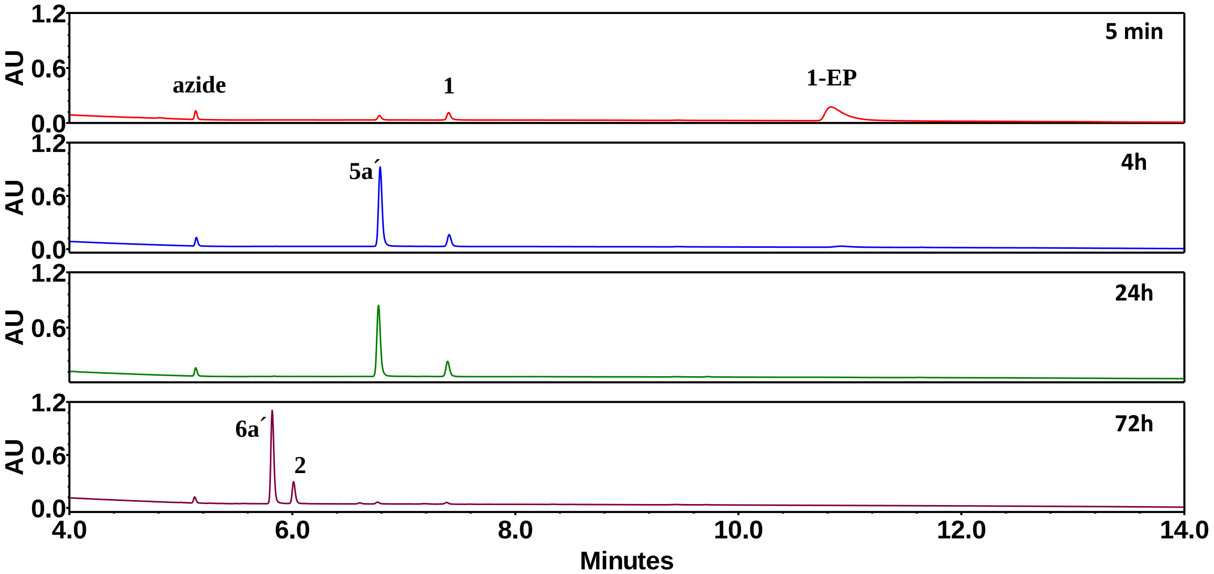
Supplementary Fig. 3. Time-dependent UPLC chromatograms of reaction between 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-YF** in 0.2 M bicarbonate buffer (pH 10.0).



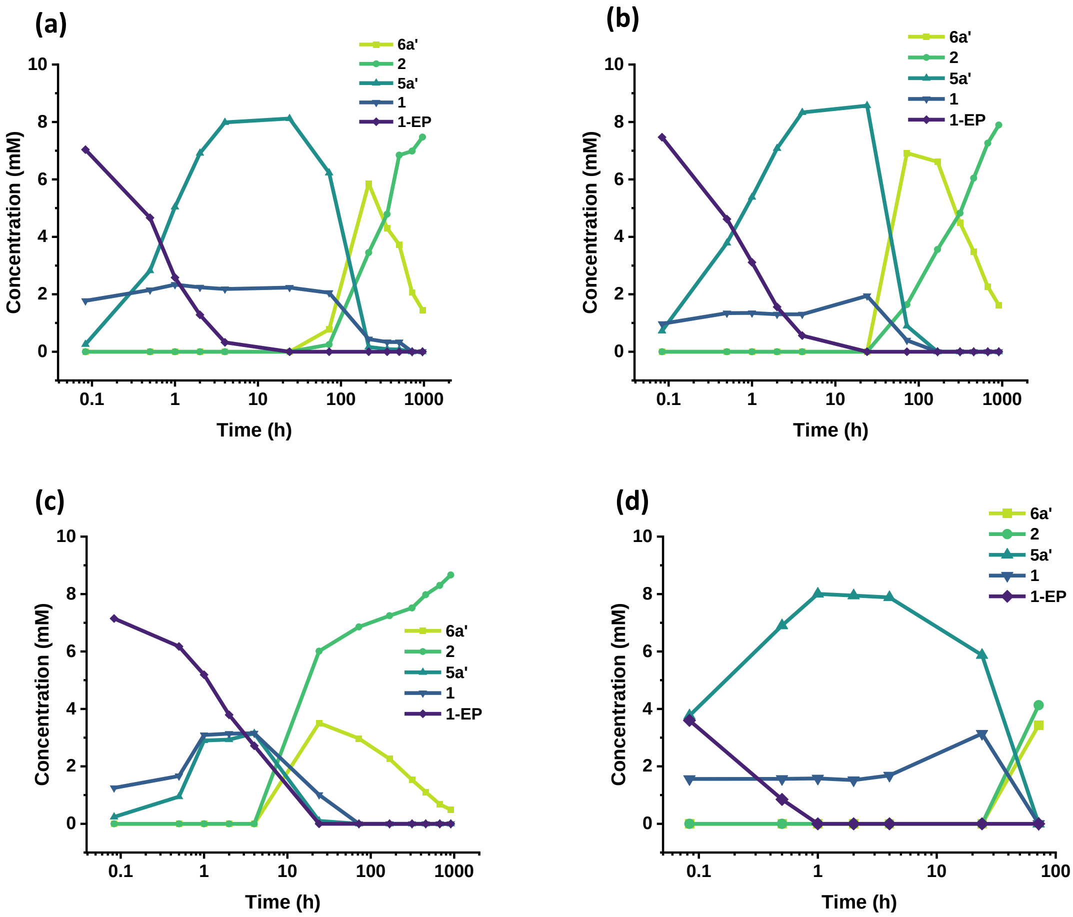
Supplementary Fig. 4. Concentration of different species from the reaction of 10 mM **1-EP**, 20 mM **azide**, and 40 mM **Ac-YF** showing the formation of **3c** and its subsequent conversion to **4c**. The reaction was carried out in 0.2 M bicarbonate buffer at pH 10.0.

3.2 Libraries from cysteine derivatives

3.2.1 Ac-C

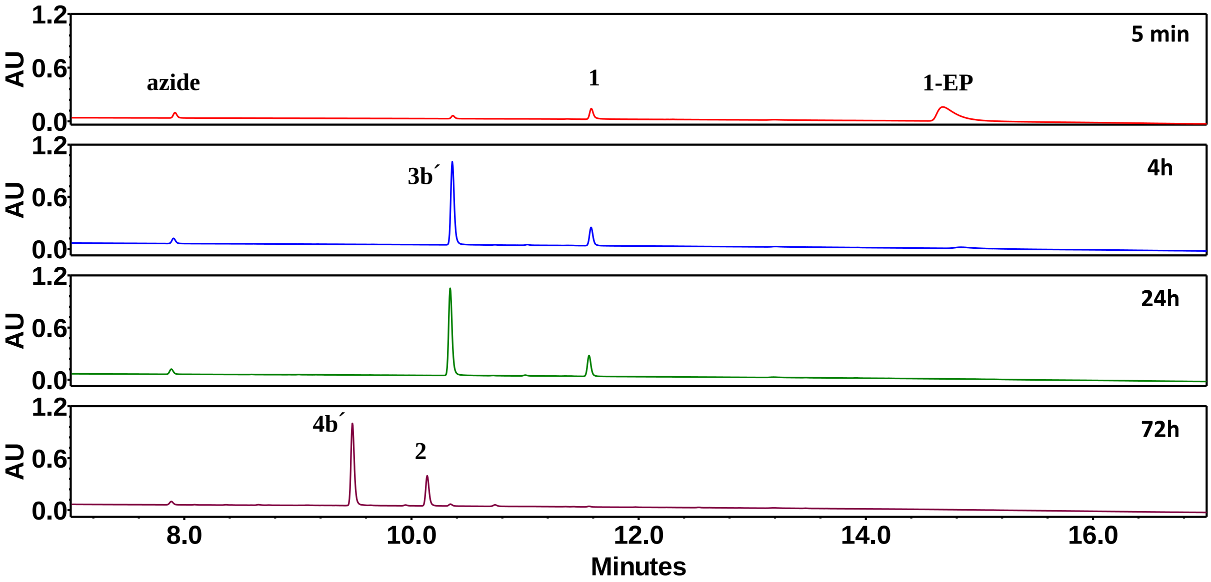


Supplementary Fig. 5. Time-dependent UPLC chromatograms of reaction between 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-C** in 0.2 M phosphate buffer (pH 8.0).

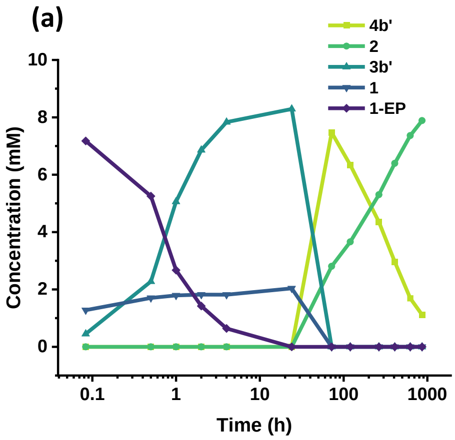


Supplementary Fig. 6. Concentration profiles of different species formed from the reaction of **1-EP** with **Ac-C** and **Azide**, showing the formation of **5a´**, and their subsequent conversion to **6a´** under varying azide and amino acid concentrations. (a) Reaction with 10 mM **1-EP**, 10 mM **azide**, and 40 mM **Ac-C**. (b) Reaction with 10 mM **1-EP**, 20 mM **Azide**, and 40 mM **Ac-C**. (c) Reaction with 10 mM **1-EP**, 40 mM **azide**, and 20 mM **Ac-C**. (d) Reaction with 10 mM **1-EP**, 20 mM **azide**, and 40 mM **Ac-C**. Reactions (a,b,c) were carried out in 0.2 M phosphate buffer at pH 8.0 and (d) was carried out in 0.2 M bicarbonate buffer at pH 10.0.

3.2.2 Ac-CD

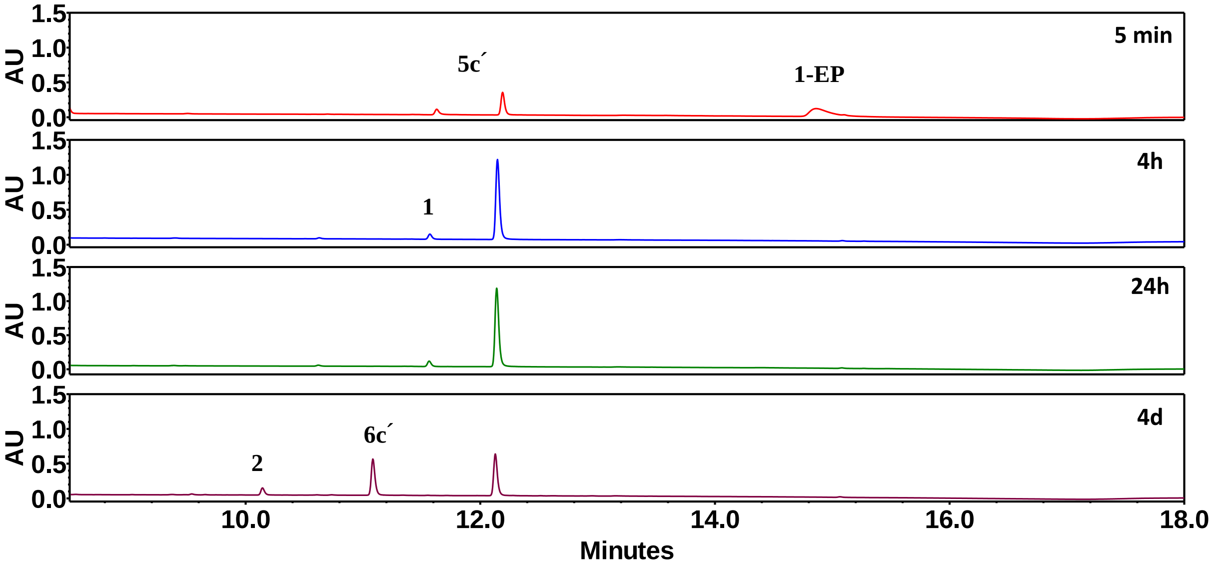


Supplementary Fig. 7. Time-dependent UPLC chromatograms of reaction between 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-CD** in 0.2 M phosphate buffer (pH 8.0).

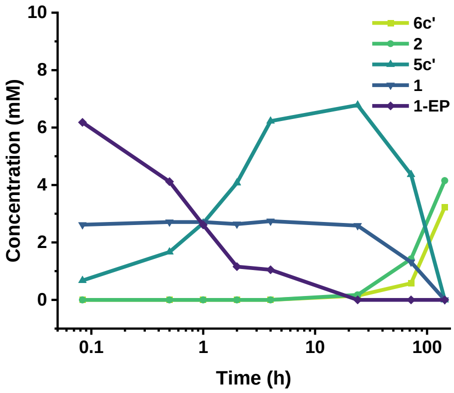


Supplementary Fig. 8. Concentration profiles of different species formed from the reaction of **1-EP** with **Ac-CD** and **azide**, showing the formation of **3b´**, and their subsequent conversion to **4b'** under varying azide and peptide concentrations. (a) Reaction with 10 mM **1-EP**, 20 mM **azide**, and 40 mM **Ac-CD**. (b) Reaction with 10 mM **1-EP**, 40 mM **azide**, and 20 mM **Ac-CD**. All reactions were carried out in 0.2 M phosphate buffer at pH 8.0.

3.2.3 Ac-CF

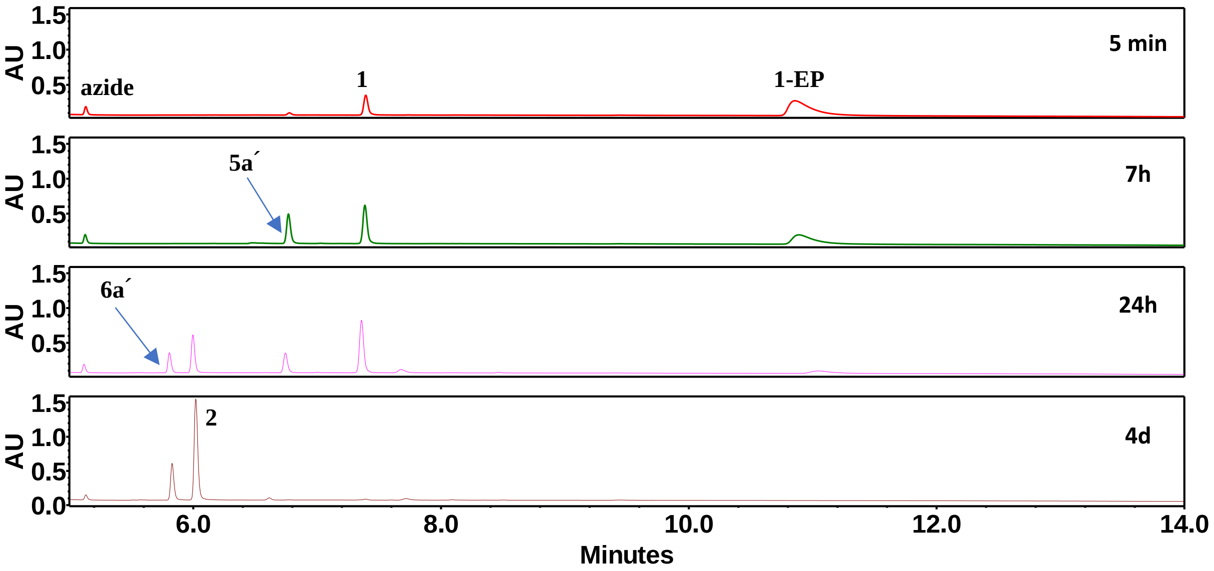


Supplementary Fig. 9. Time-dependent UPLC chromatograms of reaction between 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-CF** in 0.2 M phosphate buffer (pH 8.0).

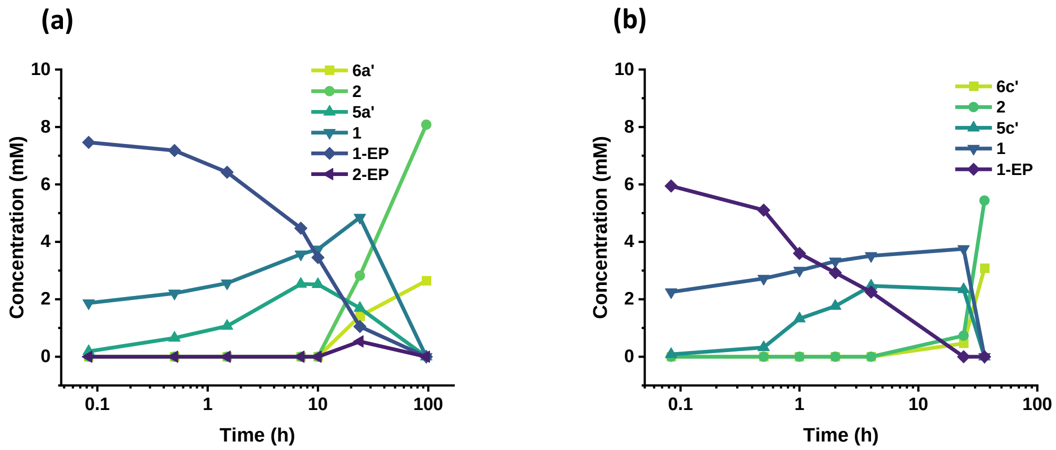


Supplementary Fig. 10. Concentration profiles of different species formed from the reaction of 10 mM **1-EP** with 40 mM **azide** and 20 mM **Ac-CF** in 0.2 M phosphate buffer (pH 8.0). The plot highlights the formation of **5c**´ and its subsequent conversion to **6c**´.

3.4 Co-Solvent experiment

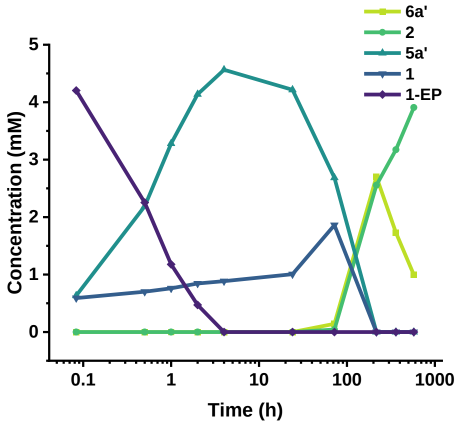


Supplementary Fig. 11. Time-dependent UPLC chromatograms of reaction between 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-C** in a 1:1 mixture of DMSO and 0.2 M phosphate buffer (pH 8.0).



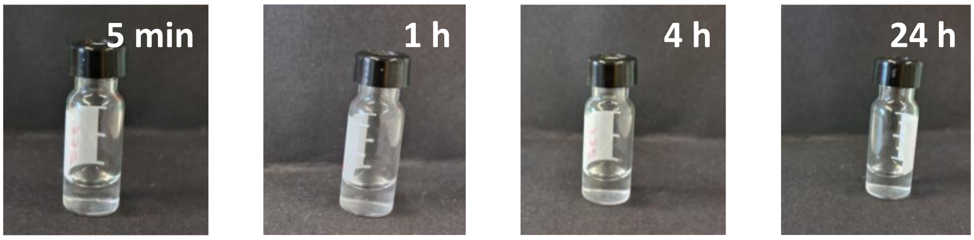
Supplementary Fig. 12. Concentration profile of the reaction of (a) 10 mM **1-EP**, 20 mM **azide**, and 40 mM **Ac-C**, showing the formation **5a´** and its subsequent conversion to **6a´** (b) 10 mM **1-EP**, 20 mM **azide**, and 40 mM **Ac-CF**, showing the formation **5c´** and its subsequent conversion to **6c´** The reactions were carried out in a 1:1 mixture of DMSO and 0.2 M phosphate buffer (pH 8.0).

3.5 Low Concentration experiment

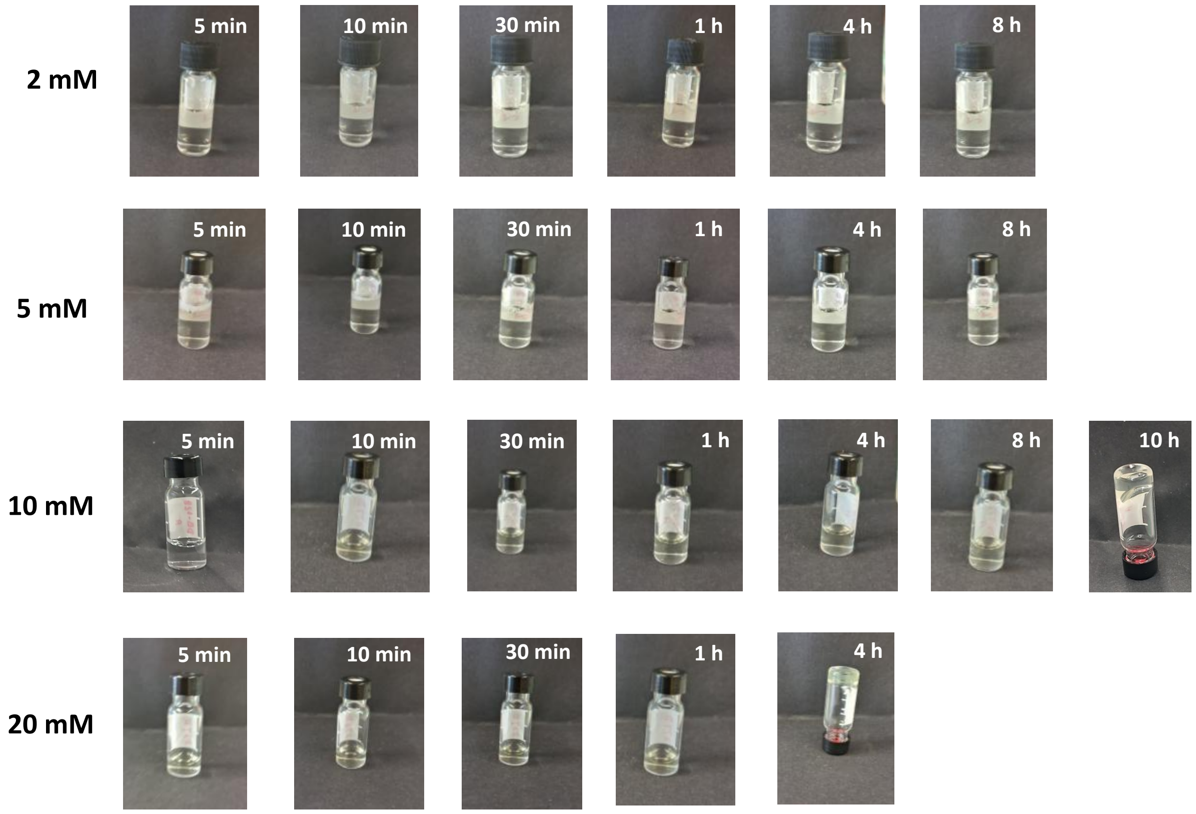


Supplementary Fig. 13. Concentration profile of the reaction of 5 mM **1-EP**, 10 mM **azide** and 20 mM **Ac-C** showing the formation of **5a´** and its subsequent conversion to **6a´.** The reaction was carried out in 0.2 M phosphate buffer at pH 8.0.

3.6 Concentration-dependent experiments and macroscopic observations

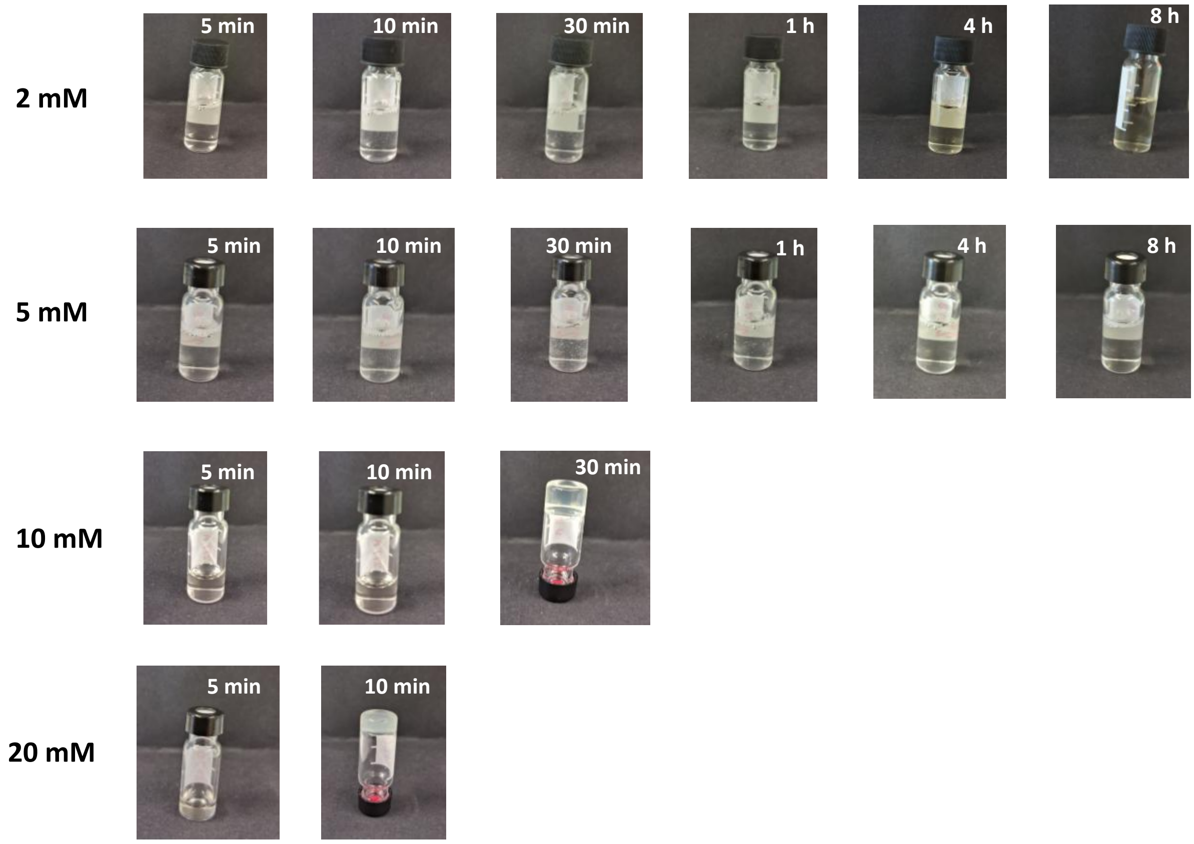


Supplementary Fig. 14. Digital photographs of 10 mM **1-EP** at various time points in 0.2 M Phosphate buffer at pH 8.0.



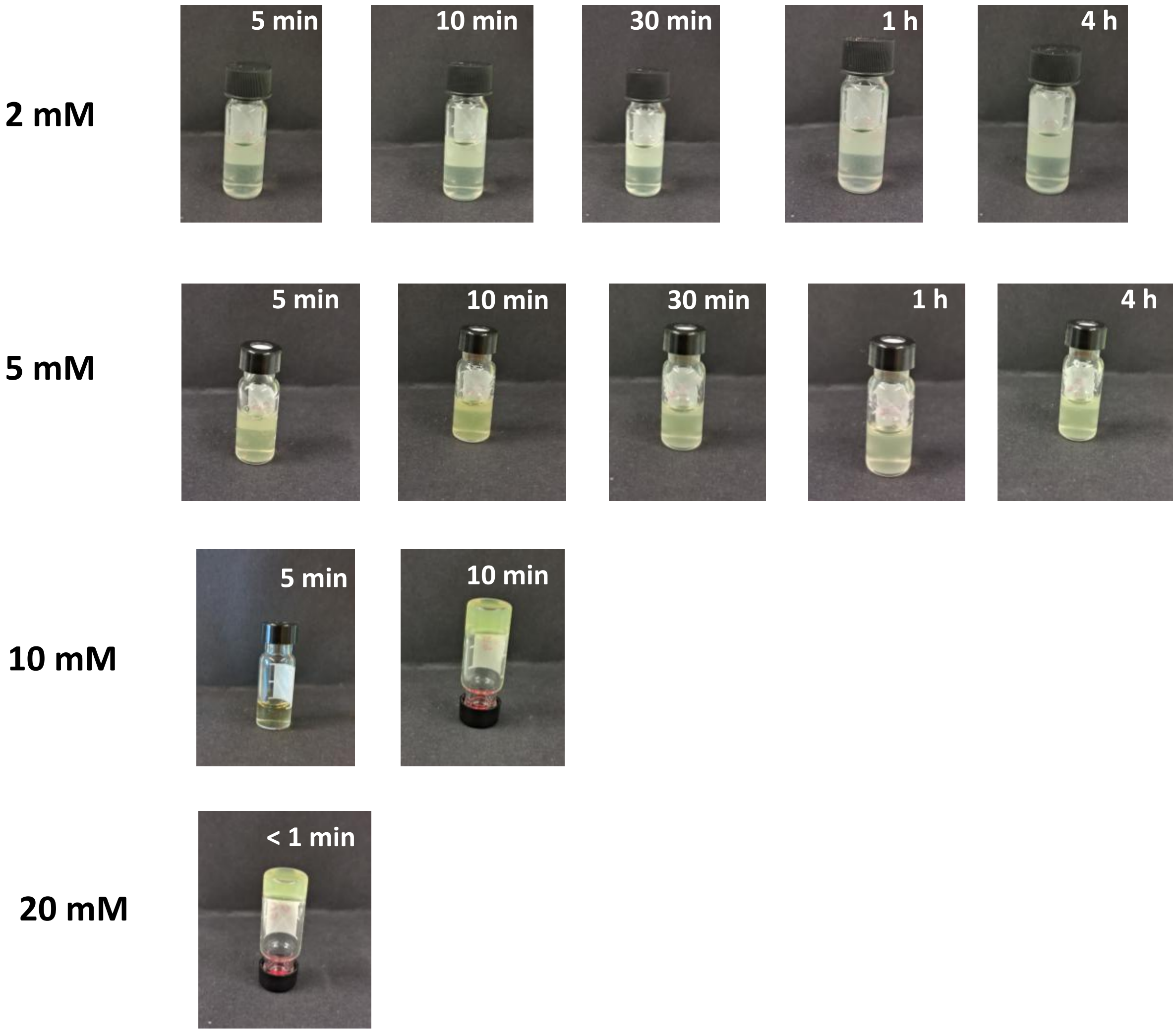
1-EP Concentration	Gelation time	Thioester (5a´) concentration during gelation
2 mM	-	-
5 mM	-	-
10 mM	10 h	8.22 mM
20 mM	4 h	8.17 mM

Supplementary Fig. 15. Concentration-dependent experiments with **1-EP**, **azide**, and **Ac-C** in a 1:2:4 ratio in 0.2 M phosphate buffer (pH 8.0), along with corresponding digital photographs at different time points. Gelation times and the thioester concentrations at the respective time points are also provided.



1-EP Concentration	Gelation time	Thioester (5c´) concentration during gelation
2 mM	-	-
5 mM	-	-
10 mM	30 min	7.08 mM
20 mM	10 min	11.52 mM

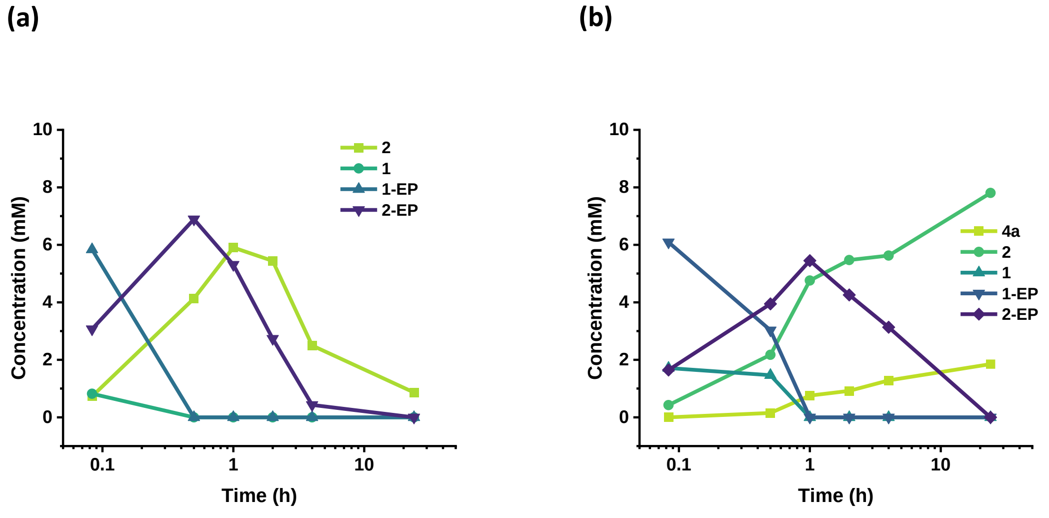
Supplementary Fig. 16. Concentration-dependent experiments with **1-EP**, **azide**, and **Ac-CF** in a 1:2:4 ratio in 0.2 M phosphate buffer (pH 8.0), along with corresponding digital photographs at different time points. Gelation times and the thioester concentrations at the respective time points are also provided.



1-EP Concentration	Gelation time	Ester (3c) concentration during gelation
2 mM	-	-
5 mM	-	-
10 mM	10 min	3.11 mM
20 mM	<1 min	3.96 mM

Supplementary Fig. 17. Concentration-dependent experiments with **1-EP**, **azide**, and **Ac-YF** in a 1:2:4 ratio in 0.2 M bicarbonate buffer (pH 10.0), along with corresponding digital photographs at different time points. Gelation times and the ester concentrations at the respective time points are also provided.

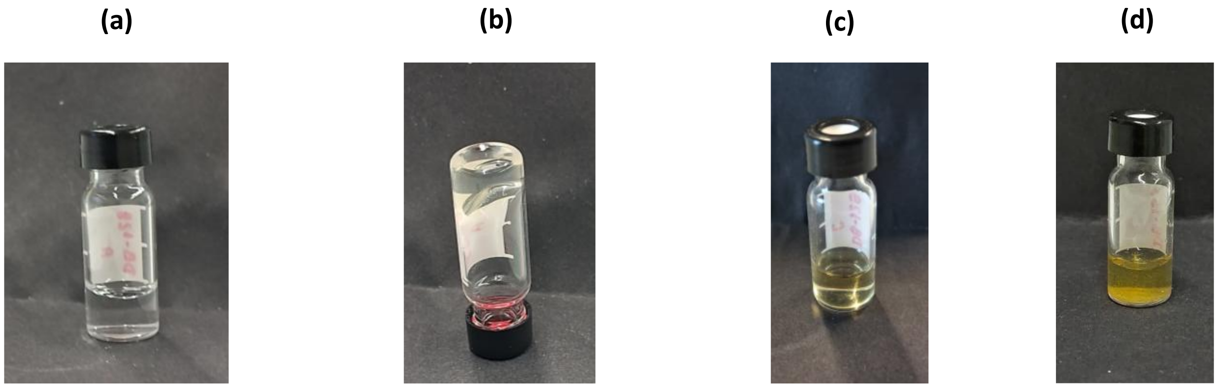
3.7 Control experiment for metal sequestration



Supplementary Fig. 18. Concentration profile of the reaction of (a) 10 mM **1-EP**, 20 mM **azide**, 40 mM **Ac-C,** 45 mM CuSO4.5H_2O and 90 mM sodium ascorbate showing the formation of **2** and **2-EP,** (b) 10 mM **1-EP**, 20 mM **Azide**, 40 mM **Ac-Y** and 40 mM N-Boc-Methionine showing the formation of **2, 2-EP** and **4a.** The reaction for (a) was carried out in 0.2 M phosphate buffer at pH 8.0 and for (b) in 0.2 M bicarbonate buffer at pH 10.0.

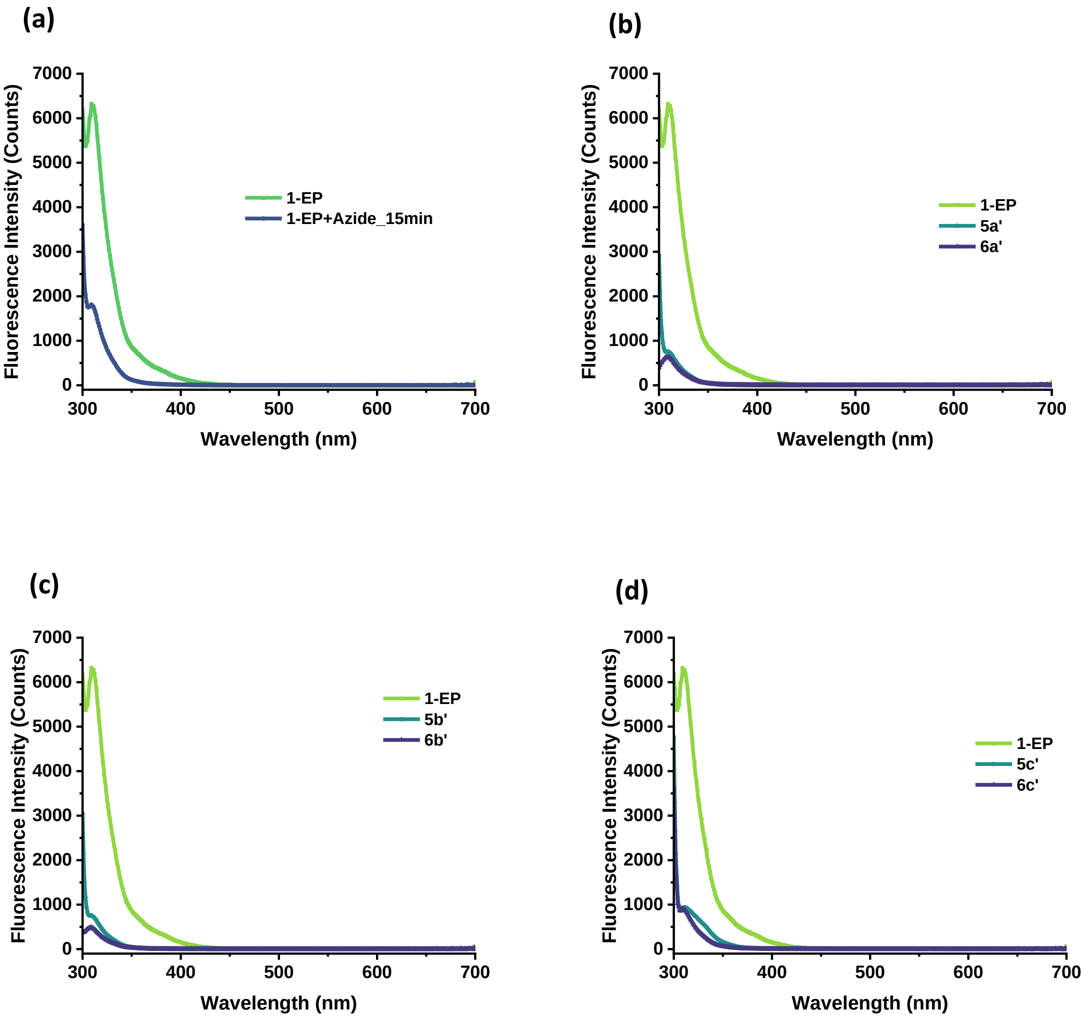
In the experiment with excess copper sulphate shown above (Figure S18b), after 2 h we have observed area loss in UPLC for **2**. Macroscopically, we have observed the formation of chunks (image attached below) precipitating out from the reaction mixture, suggesting that copper being used in excess may have chelated with the triazole of the click products formed during the course of reaction, eventually precipitating out to cause this area loss in UPLC.

3.8 Digital photographs of reaction samples



Supplementary Fig. 19. Macroscopic appearances of **(a)** 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-C** after 5 min. **(b)** 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-C** after 24 h. **(c)** 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-C** after 72 h **(d)** 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-Y** after 5 min.

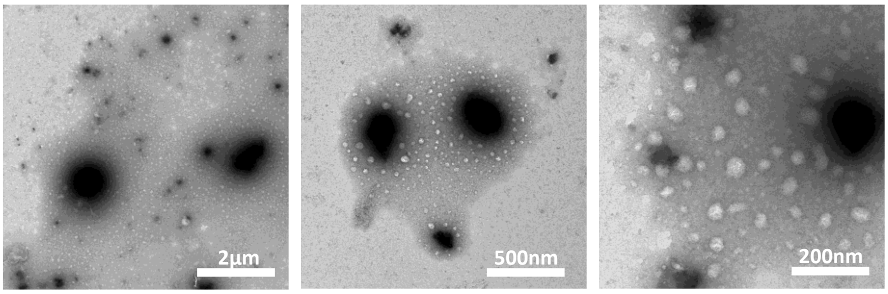
3.9 Fluorescence experiments



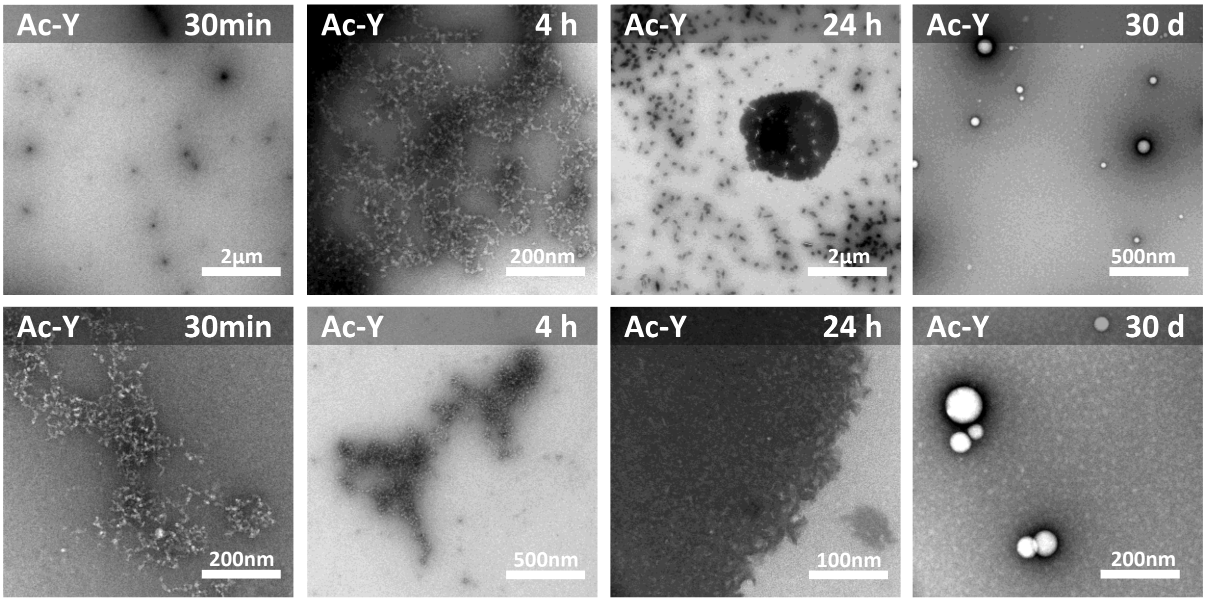
Supplementary Fig. 20. Fluorescence spectra of **(a)** 10 mM **1-EP** with 20 mM **azide. (b)** 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-C**. **(c)** 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-CD**. **(d)** 10 mM **1-EP** with 20 mM **azide** and 40 mM **Ac-CF**. All samples were prepared in 0.2 M phosphate buffer (pH 8.0).

3.10 Microscopy images

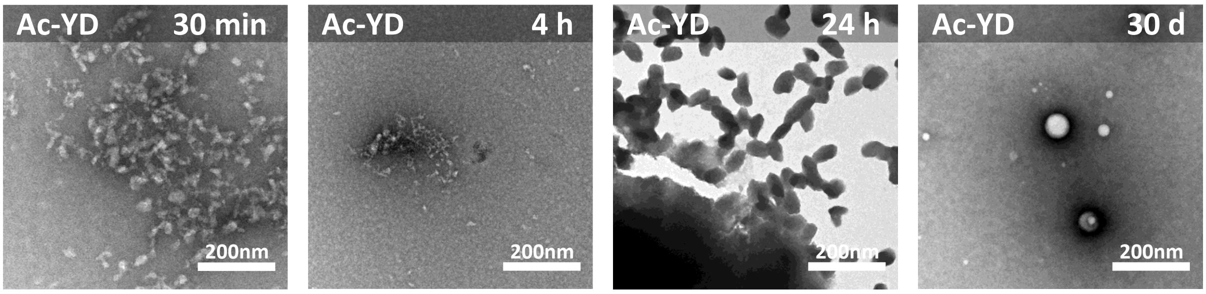
3.10.1 TEM



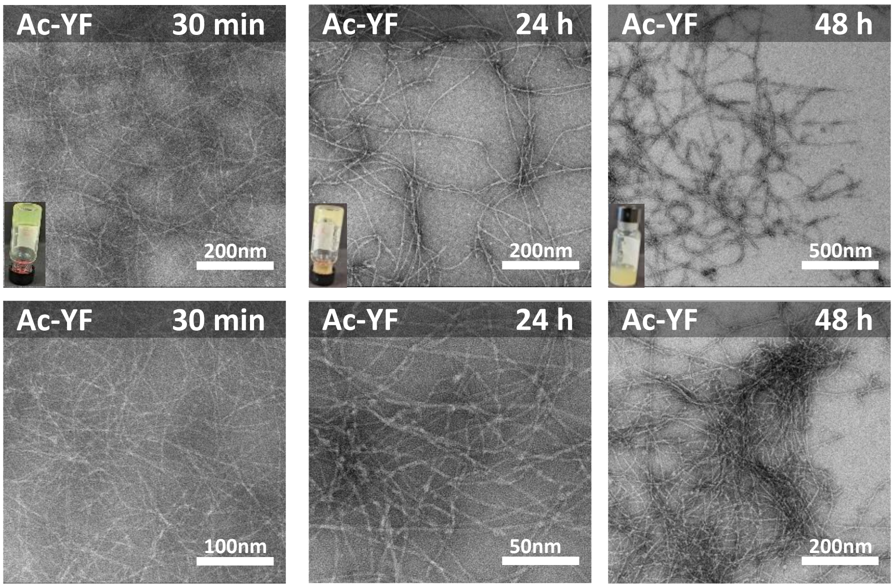
Supplementary Fig. 21. TEM images of assemblies formed from **1-EP** (10mM) in 0.2 M phosphate buffer (pH 8.0) measured at 30 min of the reaction.



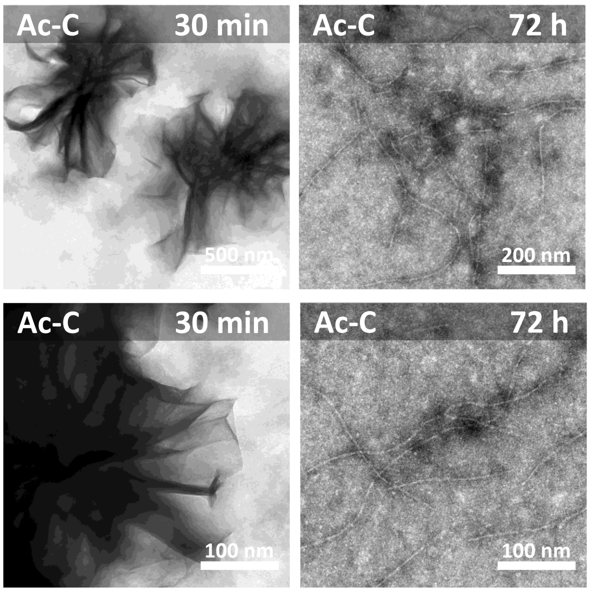
Supplementary Fig. 22. TEM images of assemblies formed from 10 mM **1-EP**, 20 mM **azide** and 40 mM **Ac-Y** in 0.2 M bicarbonate buffer at pH 10.0 at different time points.



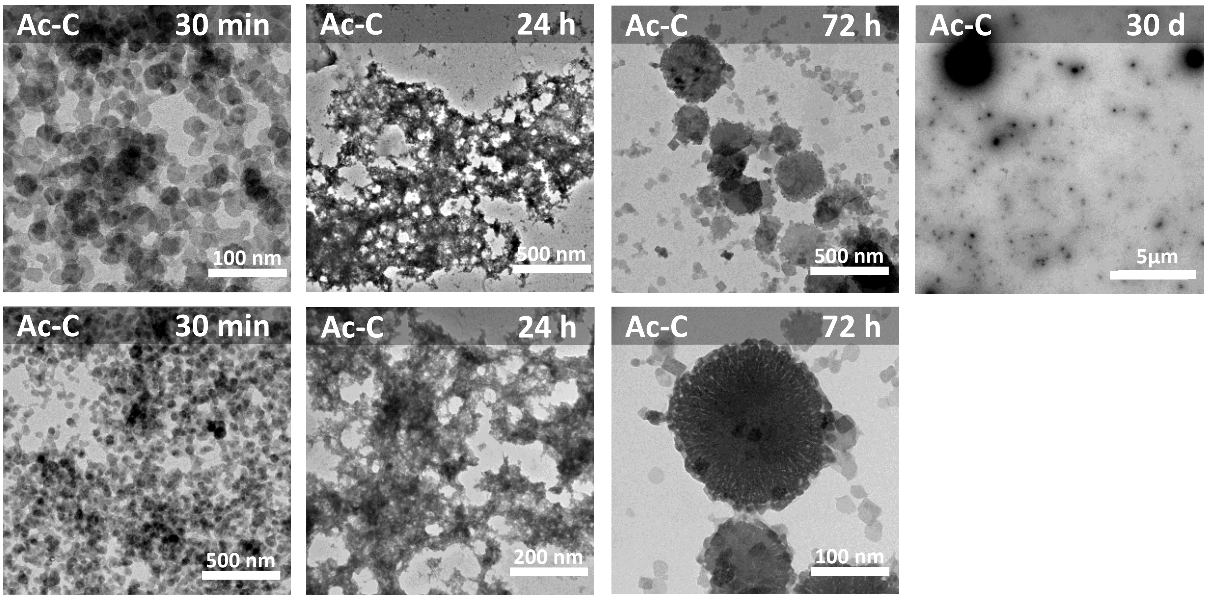
Supplementary Fig. 23. TEM images of assemblies formed from 10 mM **1-EP**, 20 mM **azide** and 40 mM **Ac-YD** in 0.2 M bicarbonate buffer at pH 10.0 at different time points.



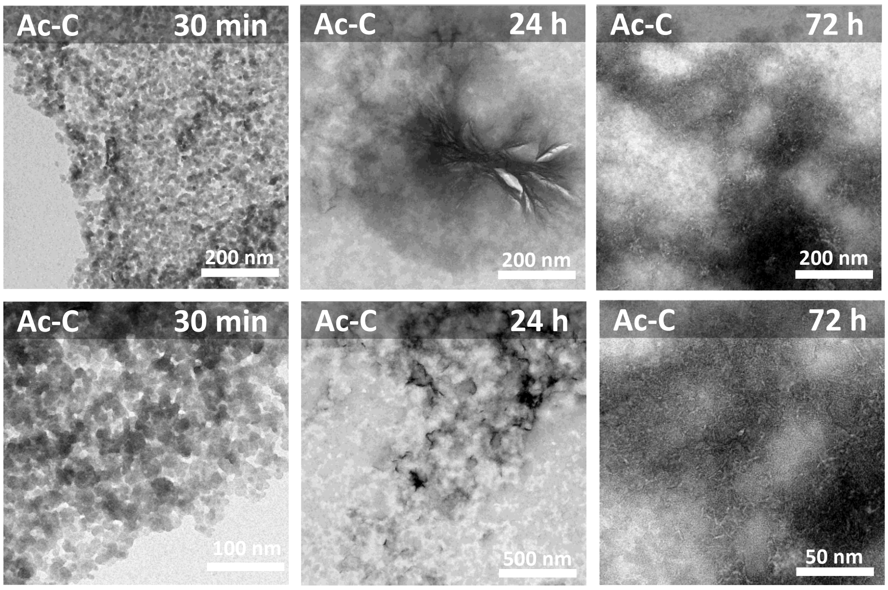
Supplementary Fig. 24. TEM images of assemblies formed from 10 mM **1-EP**, 20 mM **azide** and 40 mM **Ac-YF** in 0.2 M bicarbonate buffer at pH 10.0 at different time points.



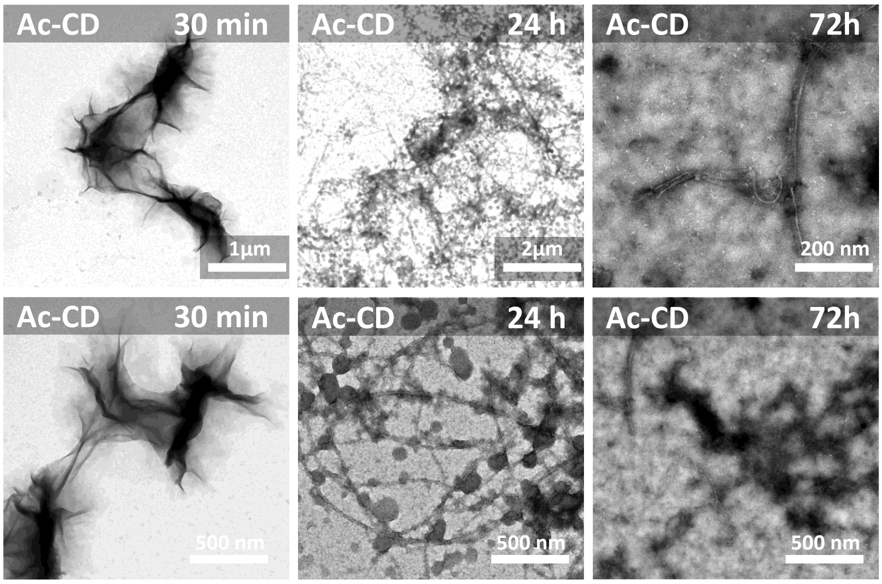
Supplementary Fig. 25. TEM images of assemblies formed from 10 mM **1-EP,** 10 mM **azide** and 40 mM **Ac-C** in 0.2 M phosphate buffer at pH 8.0 at different time points.



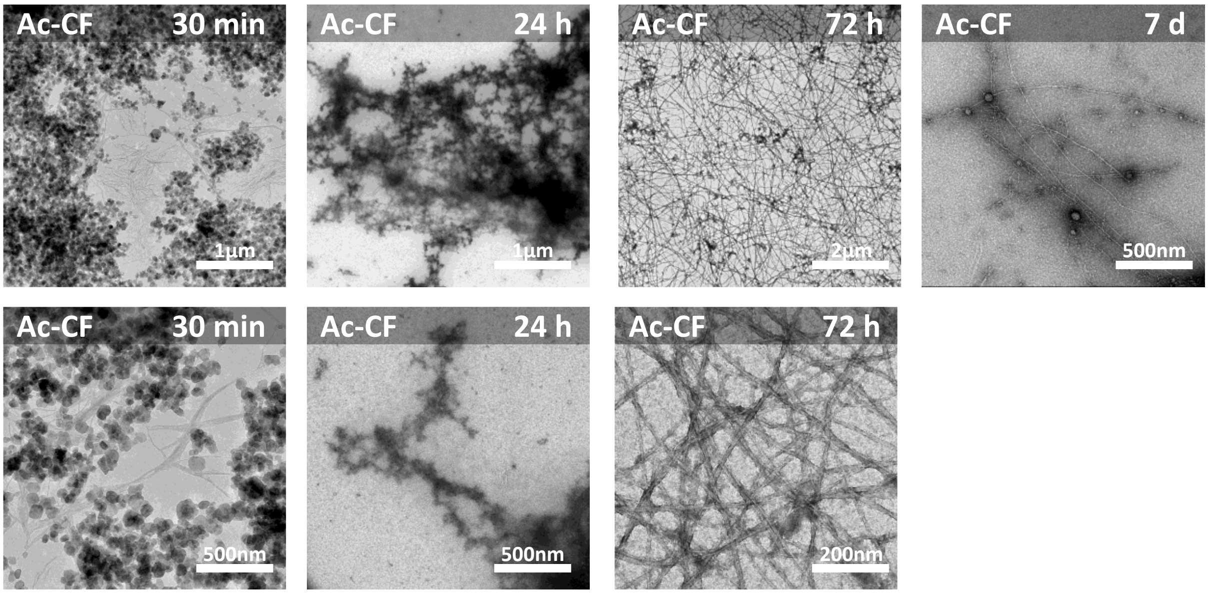
Supplementary Fig. 26. TEM images of assemblies formed from 10 mM **1-EP**, 20 mM **azide** and 40 mM **Ac-C** in 0.2 M phosphate buffer at pH 8.0 at different time points.



Supplementary Fig. 27. TEM images of assemblies formed from 10 mM **1-EP**, 40 mM **azide** and 20 mM **Ac-C** in 0.2 M phosphate buffer at pH 8.0 at different time points.

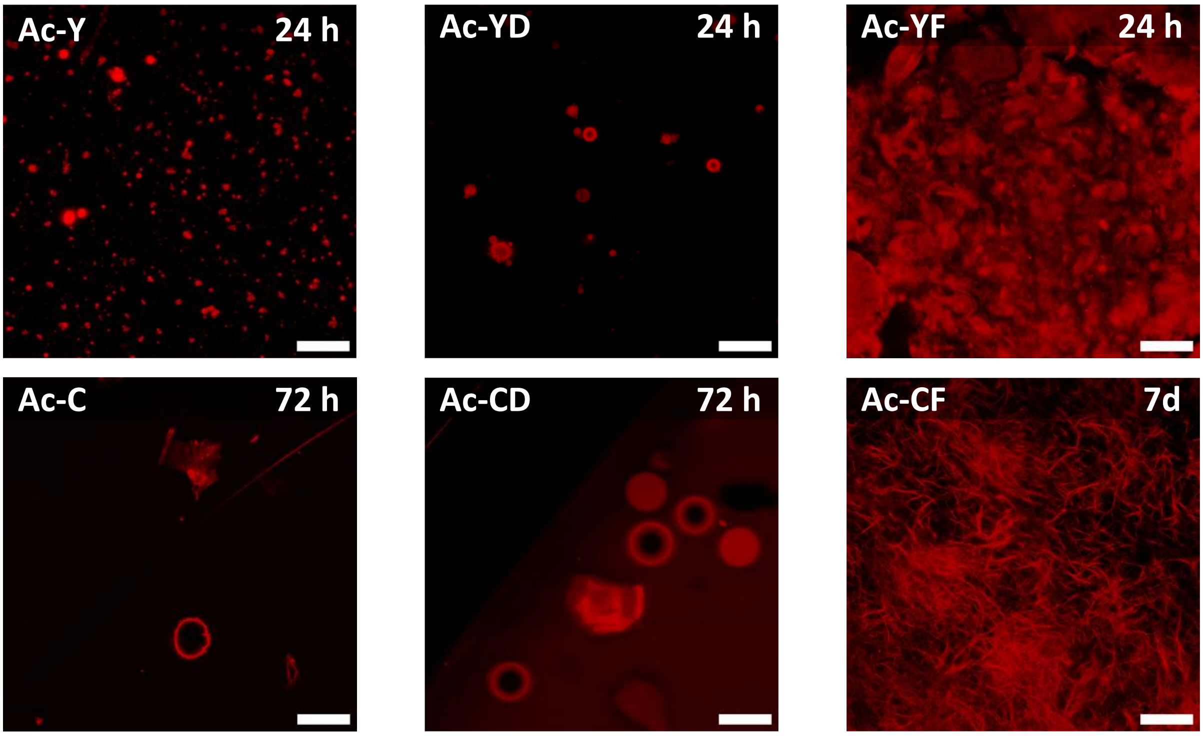


Supplementary Fig. 28. TEM images of assemblies formed from 10 mM **1-EP**, 20 mM **azide** and 40 mM **Ac-CD** in 0.2 M phosphate buffer at pH 8.0 at different time points.



Supplementary Fig. 29. TEM images of assemblies formed from 10 mM **1-EP**, 20 mM **azide** and 40 mM **Ac-CF** in 0.2 M phosphate buffer at pH 8.0 at different time points.

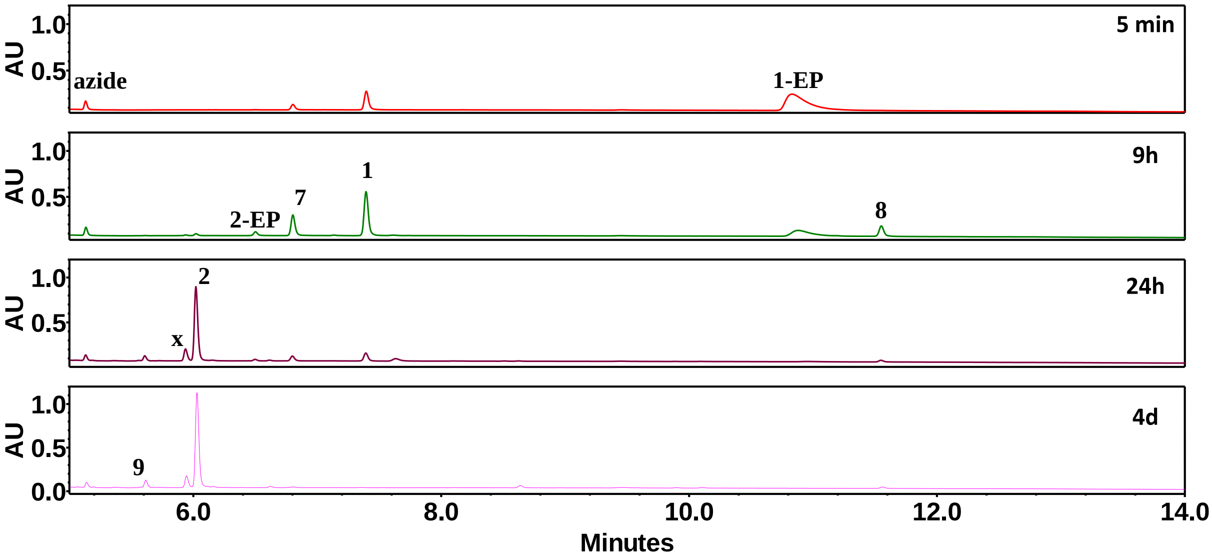
3.10.2 Confocal images



Supplementary Fig. 30. Confocal images of assemblies formed from 10 mM **1-EP**, 20 mM **azide** and 40 mM **peptidic nucleophiles** in 0.2 M phosphate buffer (pH 8.0) and 0.2 M bicarbonate buffer (pH 10.0) for cysteine and tyrosine-containing residues respectively. The scale bar for all images is 20 µm.

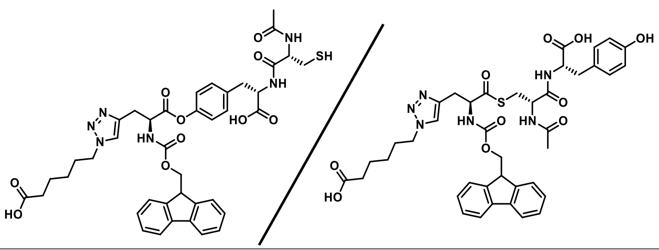
3.11 Competition experiments for multiple acyl transfer and click reactions

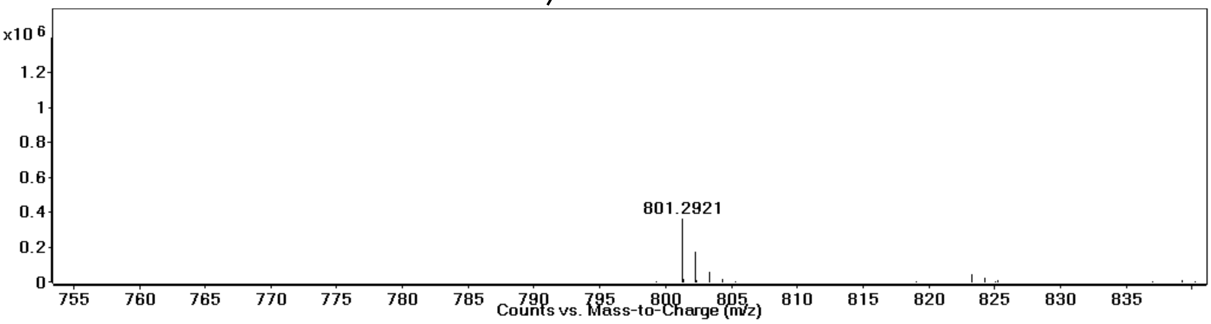
3.11.1 Ac-CY



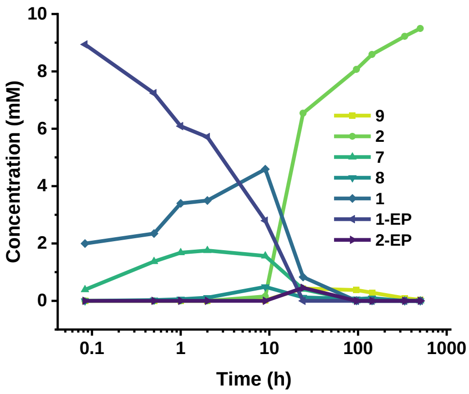
Supplementary Fig. 31. Time-dependent UPLC chromatograms of reaction between 10 mM **1-EP** with 20 mM **azide** and 10 mM **Ac-CY** in 0.2 M phosphate buffer at pH 8.0.

The peak termed as “**x**” can have two possible structures, as determined via mass analysis. It formed to a maximum extent of ~0.4 mM.



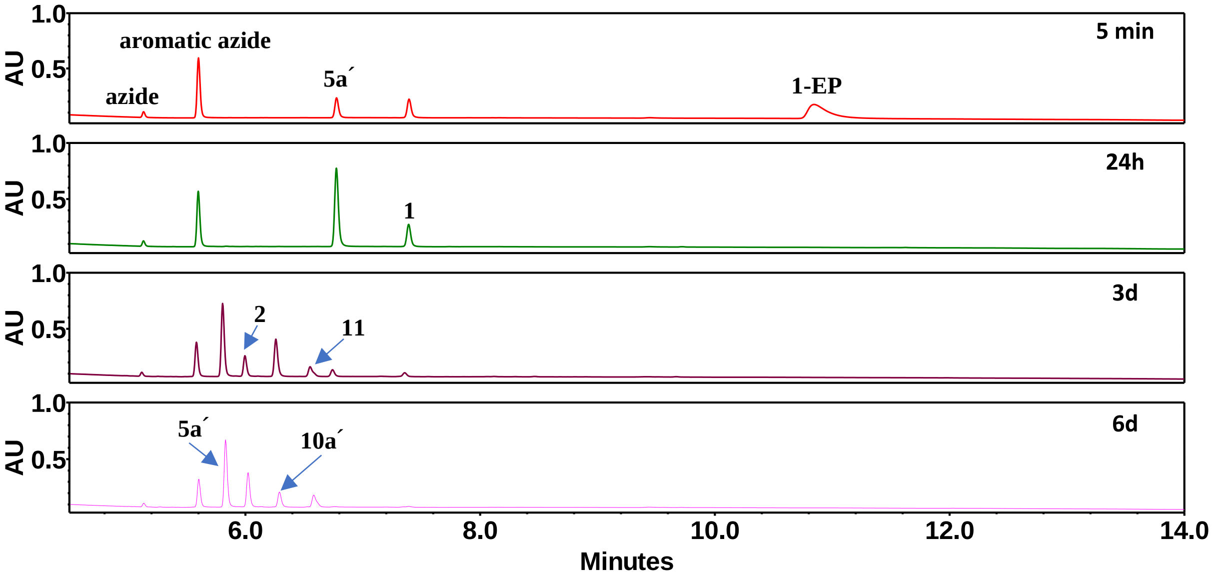


Calculated m/z $[M+H]^+$: 801.2912, Observed m/z $[M+H]^+$: 801.2921



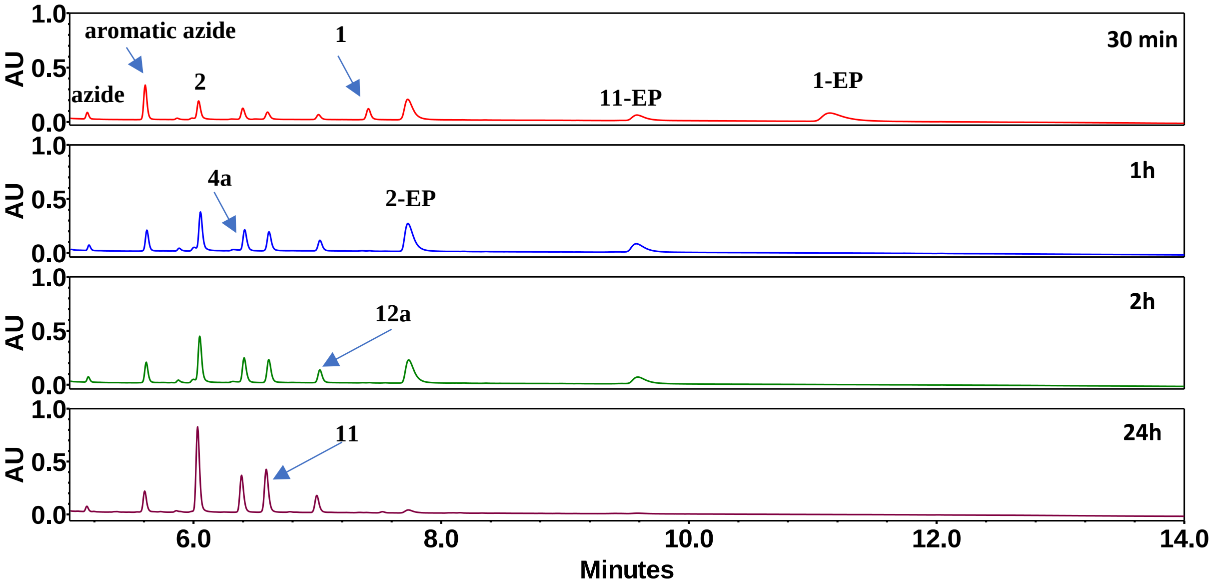
Supplementary Fig. 32. Concentration profile of the reaction of 10 mM **1-EP**, 20 mM **azide** and 10 mM **Ac-CY** showing the formation of **7**, **8** and its subsequent conversion to **9** followed by hydrolysis to give **2**. The reaction was carried out in 0.2 M phosphate buffer at pH 8.0.

3.11.2 Aromatic *vs* Aliphatic Azide with Ac-C



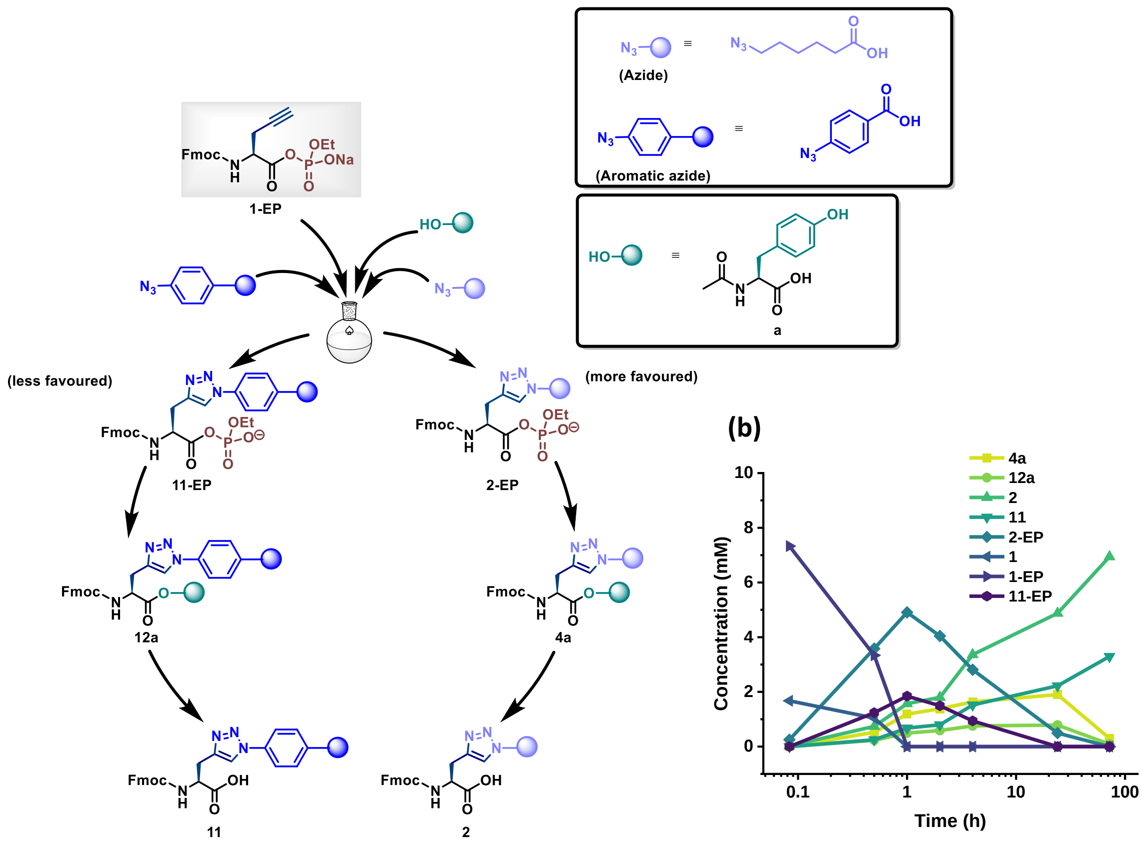
Supplementary Fig. 33. Time dependent UPLC chromatograms of reaction between 10 mM **1-EP** with 10 mM **azide**, 10 mM **aromatic azide** and 40 mM **Ac-C** in 0.2 M phosphate buffer at pH 8.0.

3.11.3 Aromatic *vs* Aliphatic Azide with Ac-Y



Supplementary Fig. 34. Time-dependent UPLC chromatograms of reaction between 10 mM **1-EP** with 10 mM **azide**, 10 mM **aromatic azide** and 40 mM **Ac-Y** in 0.2 M bicarbonate buffer at pH 10.0.

(a)



Supplementary Fig. 35. (a) Chemical structures of sequential transformations starting from **1-EP**, showing CuAAC to form **2-EP** and **11-EP**, followed by acyl transfer with **Ac-Y** to yield **4a** and **12a**. (b) Concentrations of different species formed of libraries made of 10 mM **1-EP**, 10 mM **azide**, 10 mM **aromatic azide** and 40 mM **Ac-Y** in 0.2 M bicarbonate buffer at pH 10.0.

4. References

1 Pol, M. D., Thomann, R., Thomann, Y. & Pappas, C. G. Abiotic Acyl Transfer Cascades Driven by Aminoacyl Phosphate Esters and Self-Assembly. *J. Am. Chem. Soc.* **146**, 29621-29629, (2024).